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Lee et al.

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(54) **ADAPTIVE GROUPING FOR COOPERATIVE
RADAR SENSING**

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(57) **ABSTRACT**

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Certain aspects of the present disclosure provide techniques for adaptive grouping for cooperative radar sensing. A method for wireless communications by a first network entity includes receiving, from a second network entity, first grouping information comprising an indication of a first plurality of network entities, the first grouping information associated with a first time-period; receiving, from the second network entity, second grouping information comprising an indication of a second plurality of network entities, the second grouping information associated with a second time-period; receiving a first radar signal during the first time-period; receiving a second radar signal during the second time-period; transmitting, to the second network entity, first sensing information based on the first radar signal; and transmitting, to the second network entity, second sensing information based on the second radar signal.

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(52) **U.S. Cl.**

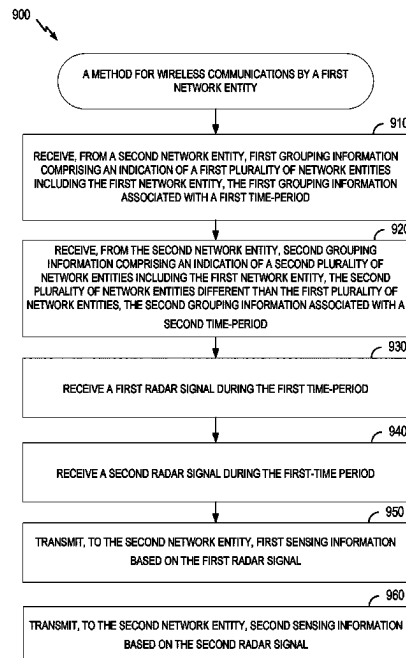
CPC **H04W 76/14** (2018.02); **H04W 16/28**
(2013.01); **H04W 76/15** (2018.02)

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None

See application file for complete search history.

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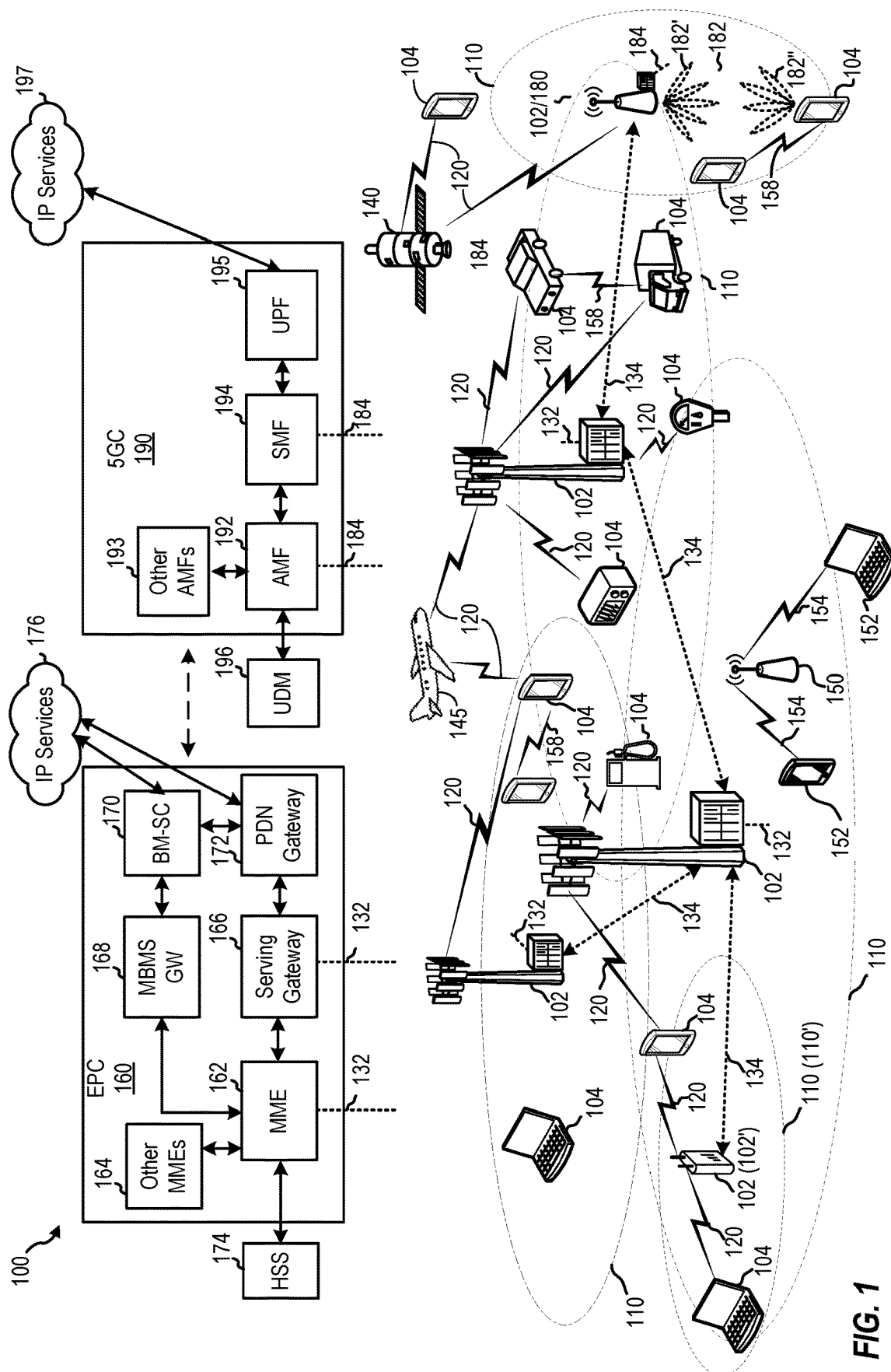
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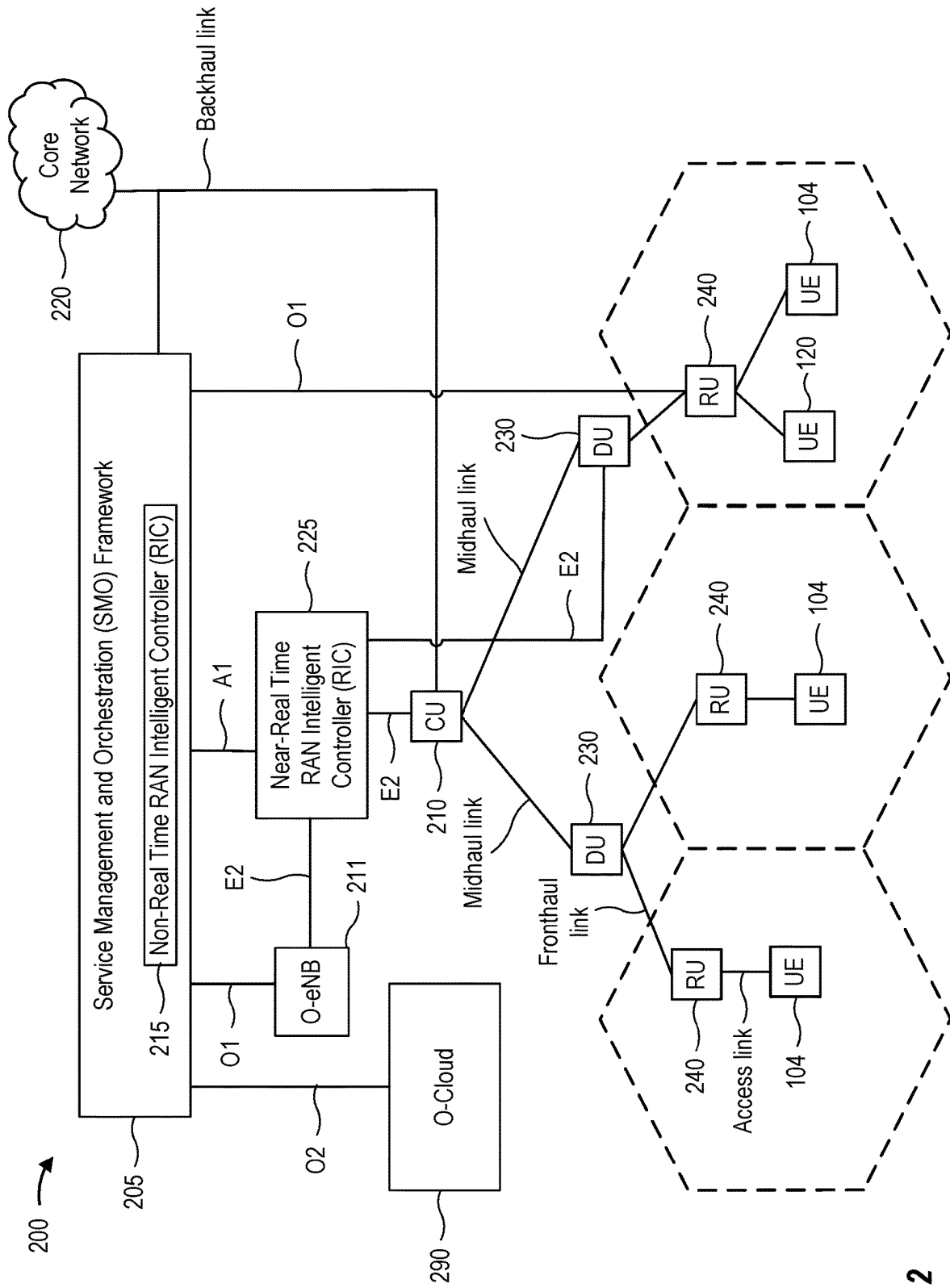


FIG. 2

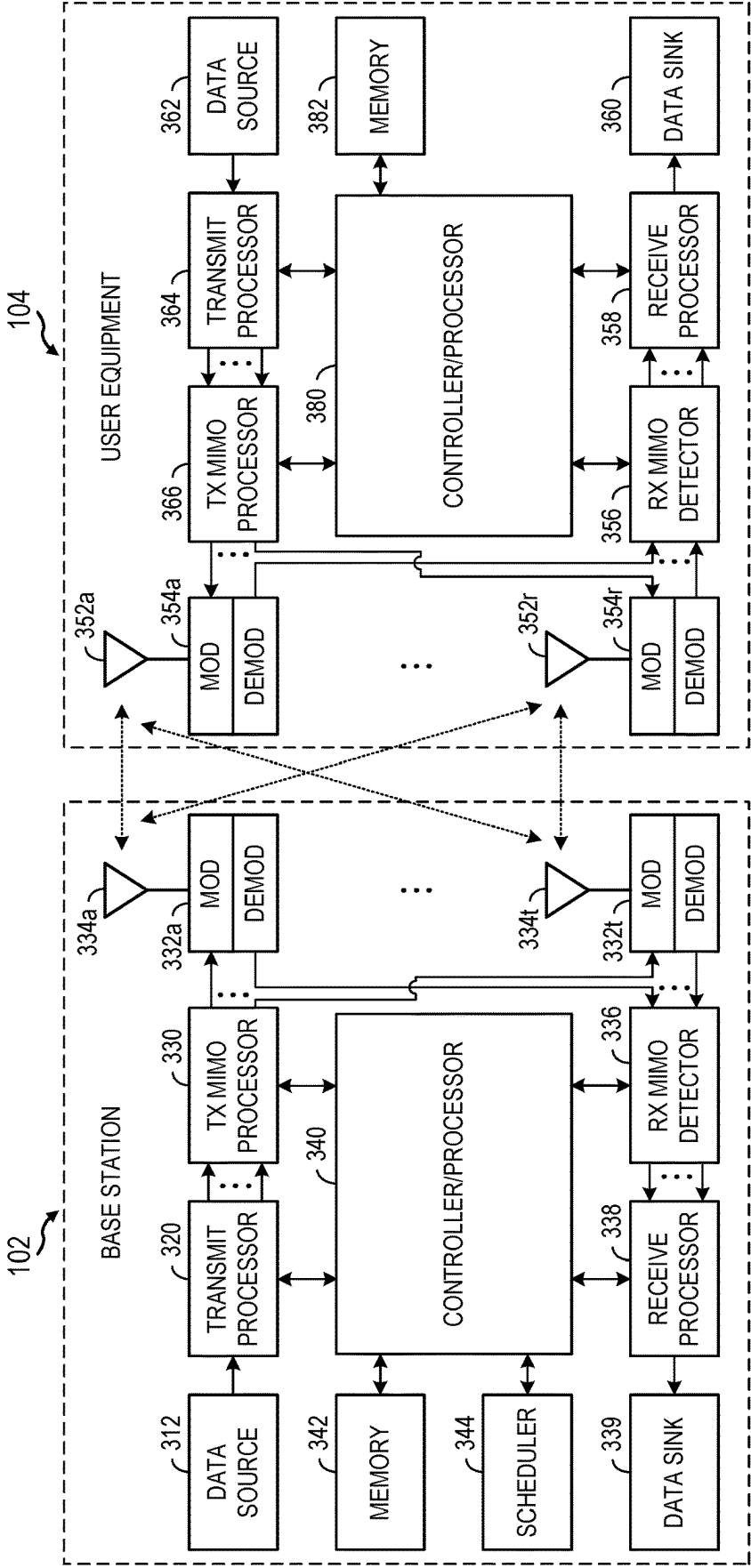
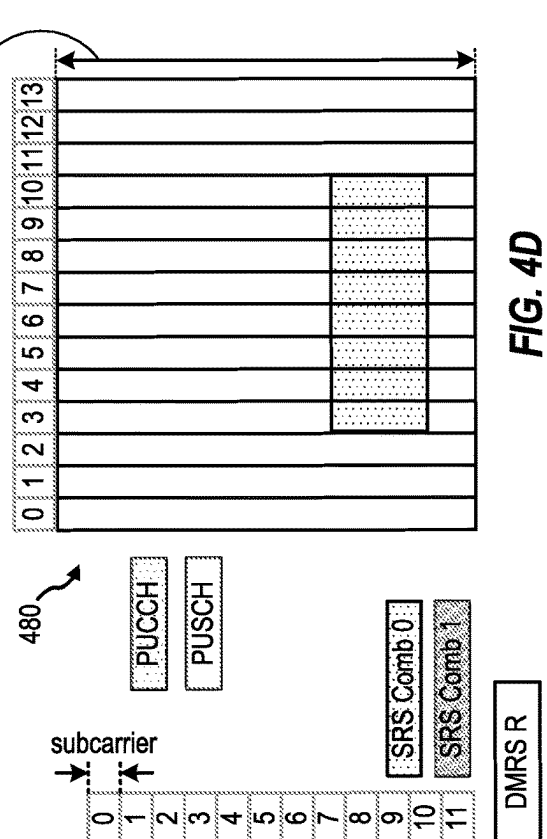
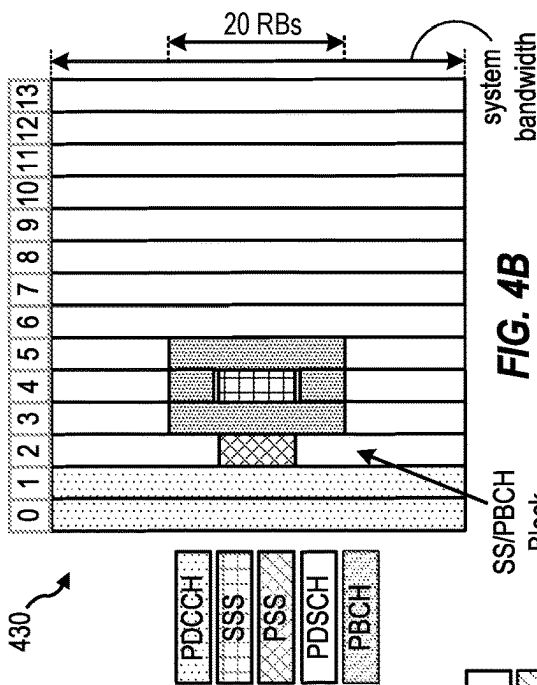
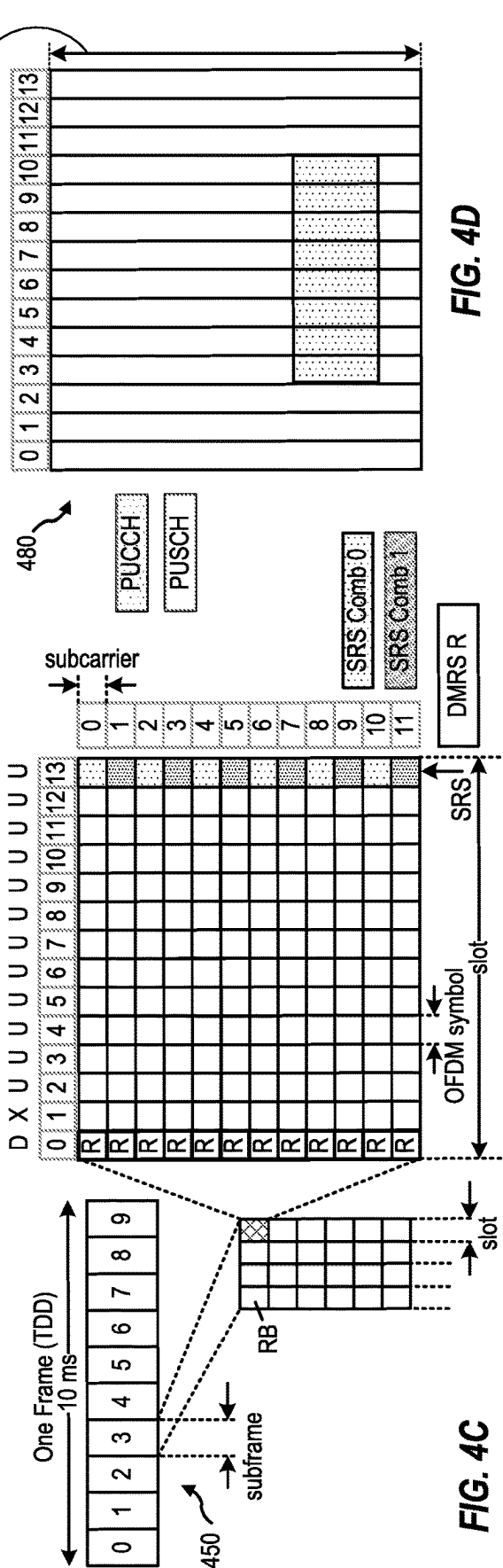
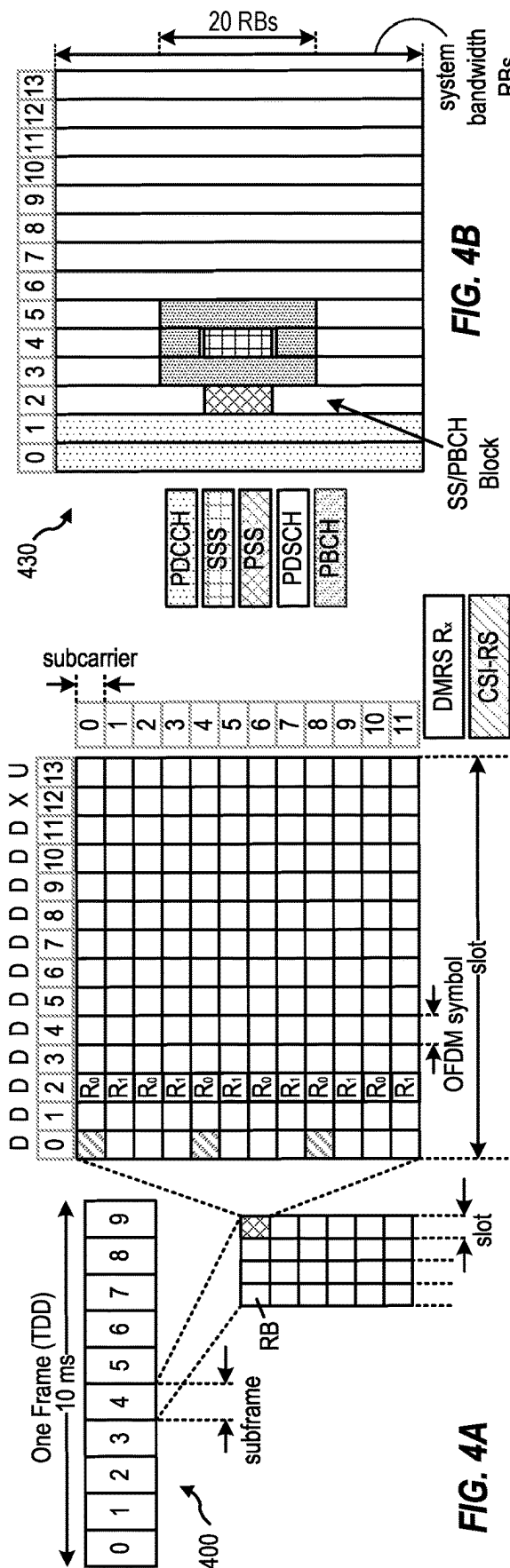


FIG. 3



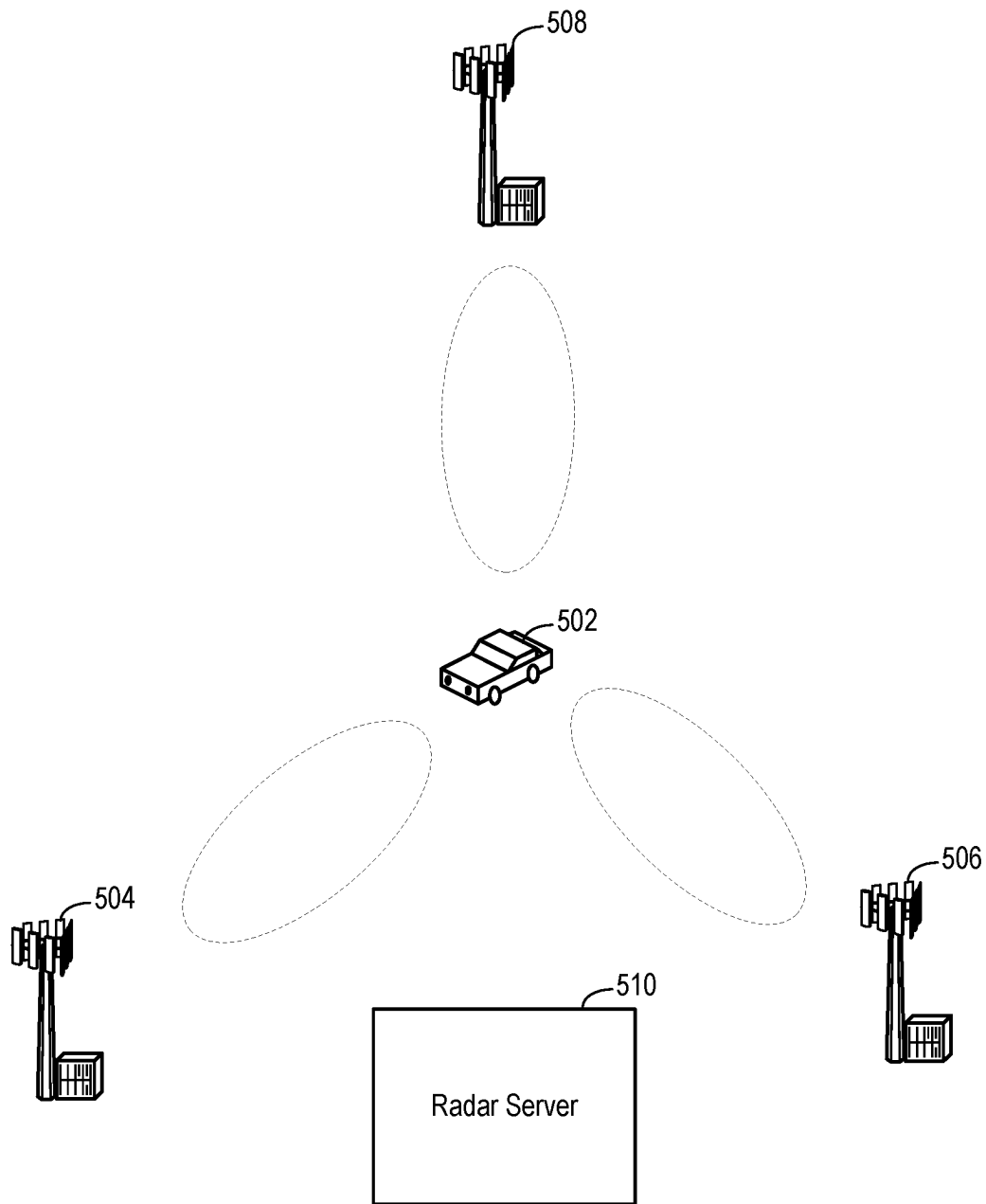


FIG. 5

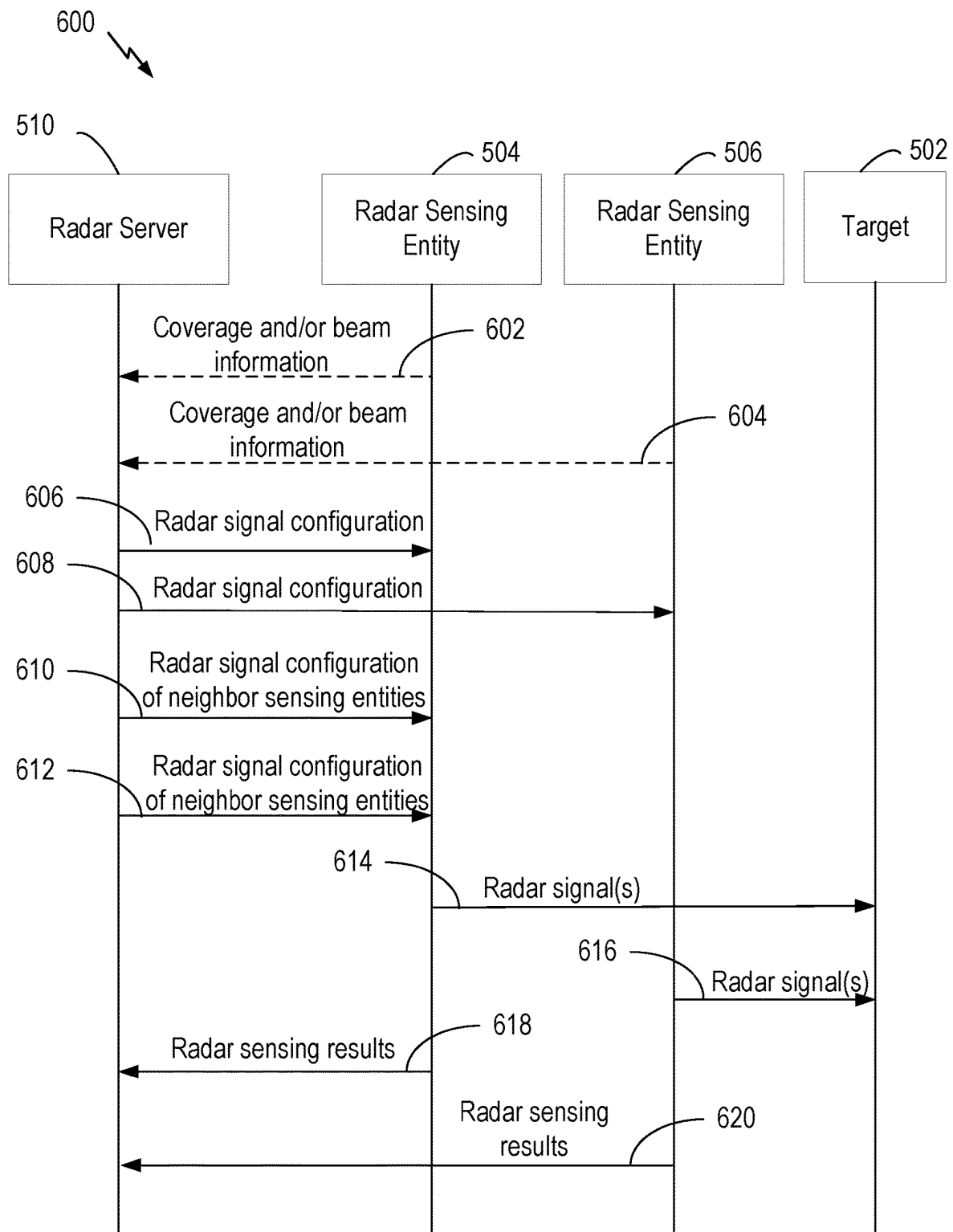


FIG. 6

700A

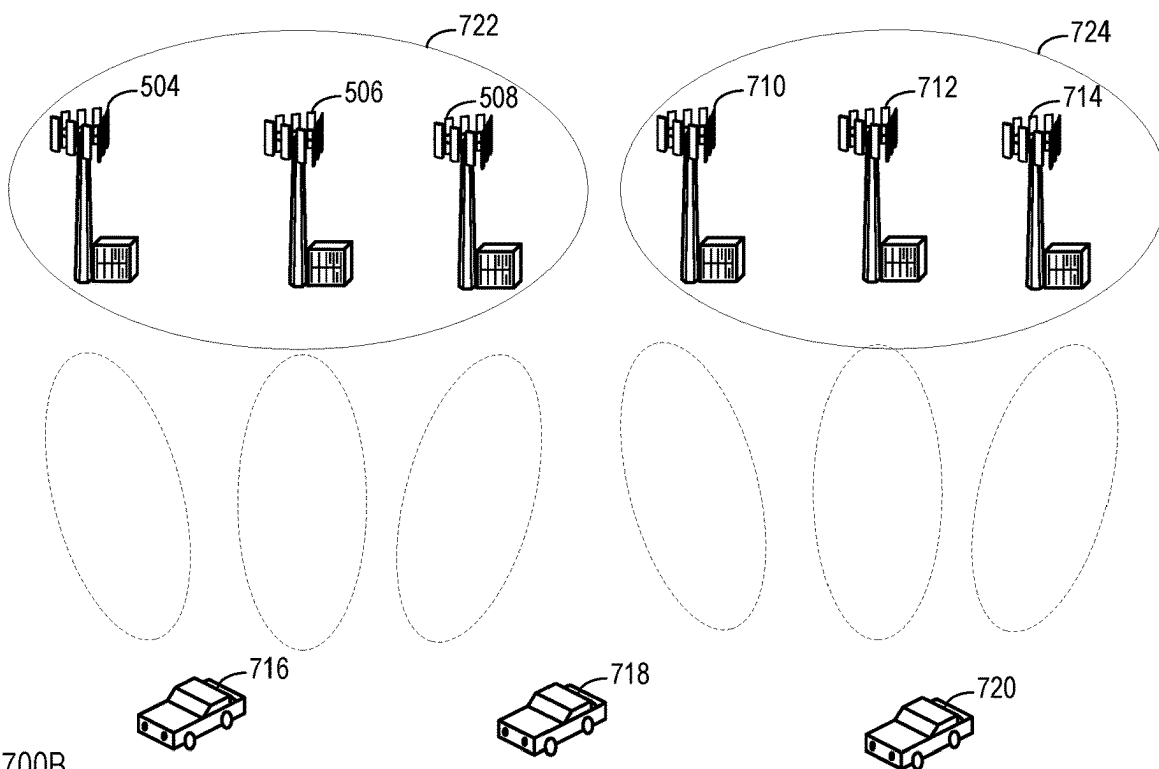


FIG. 7A

700B

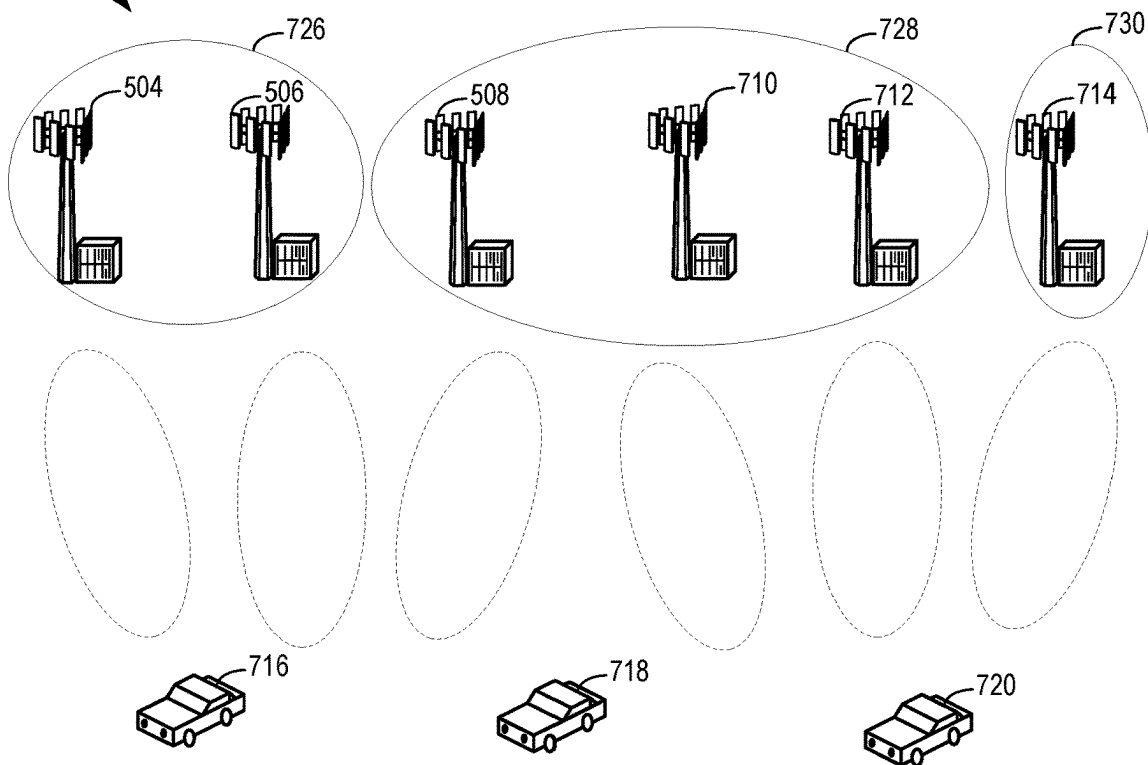
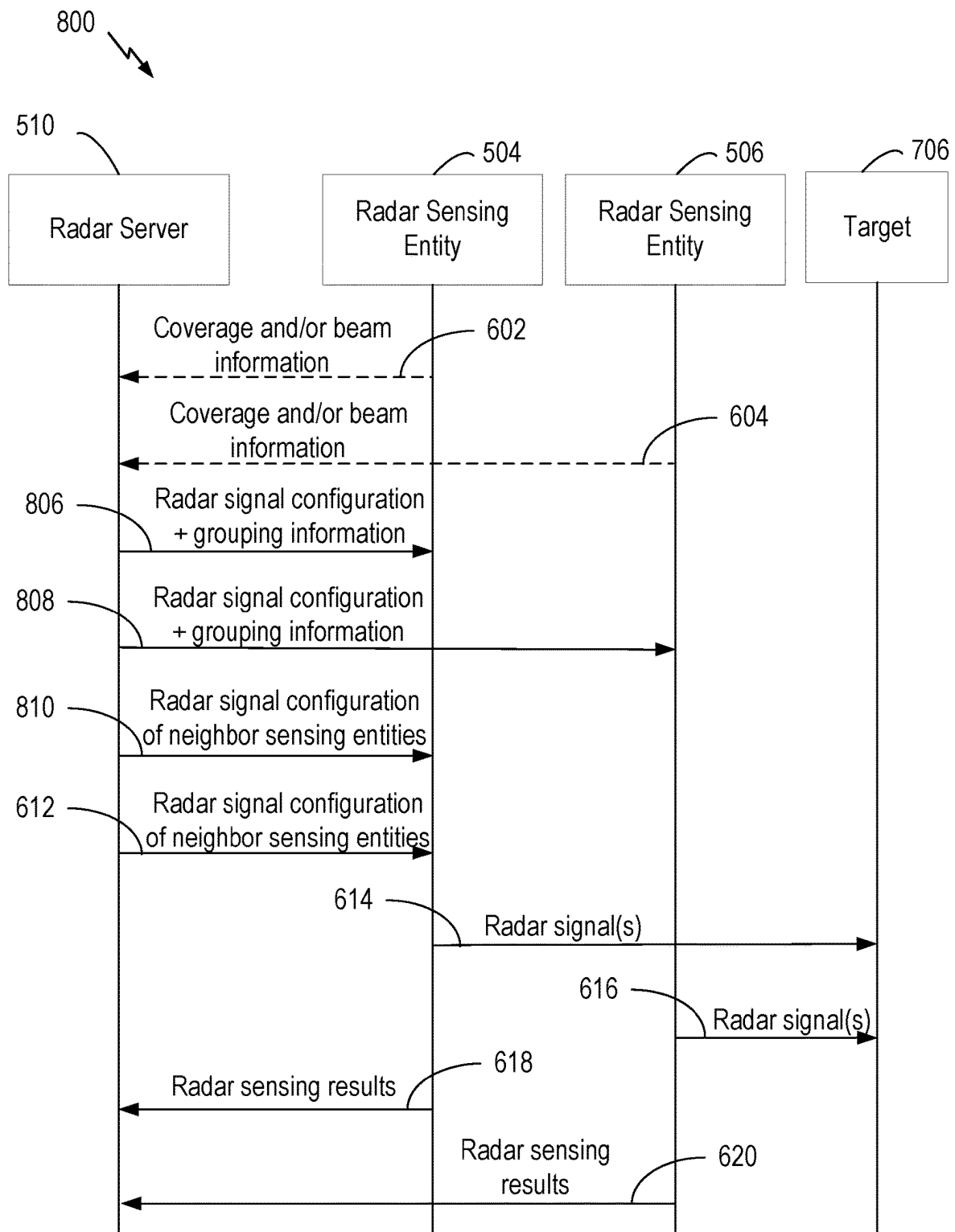
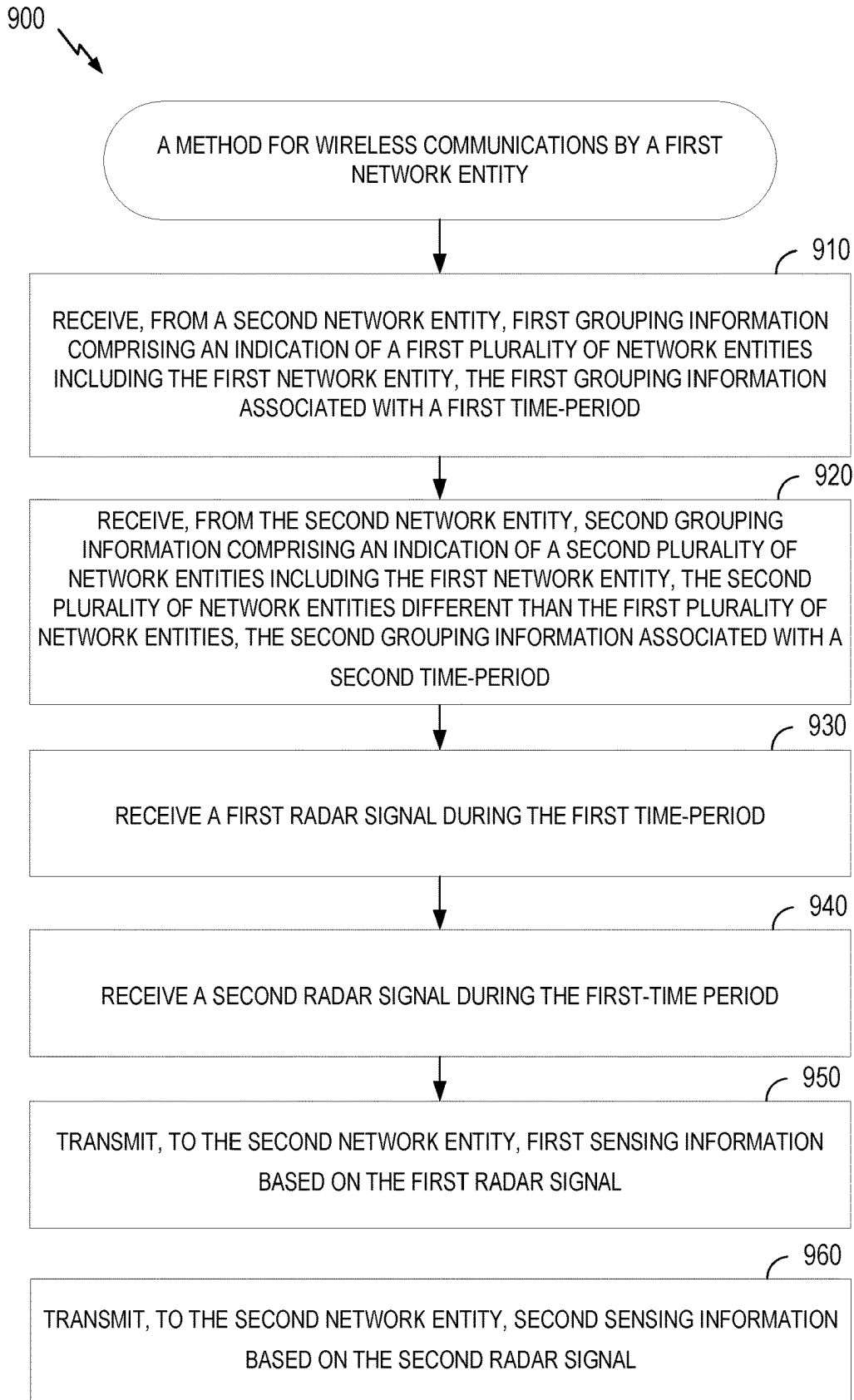
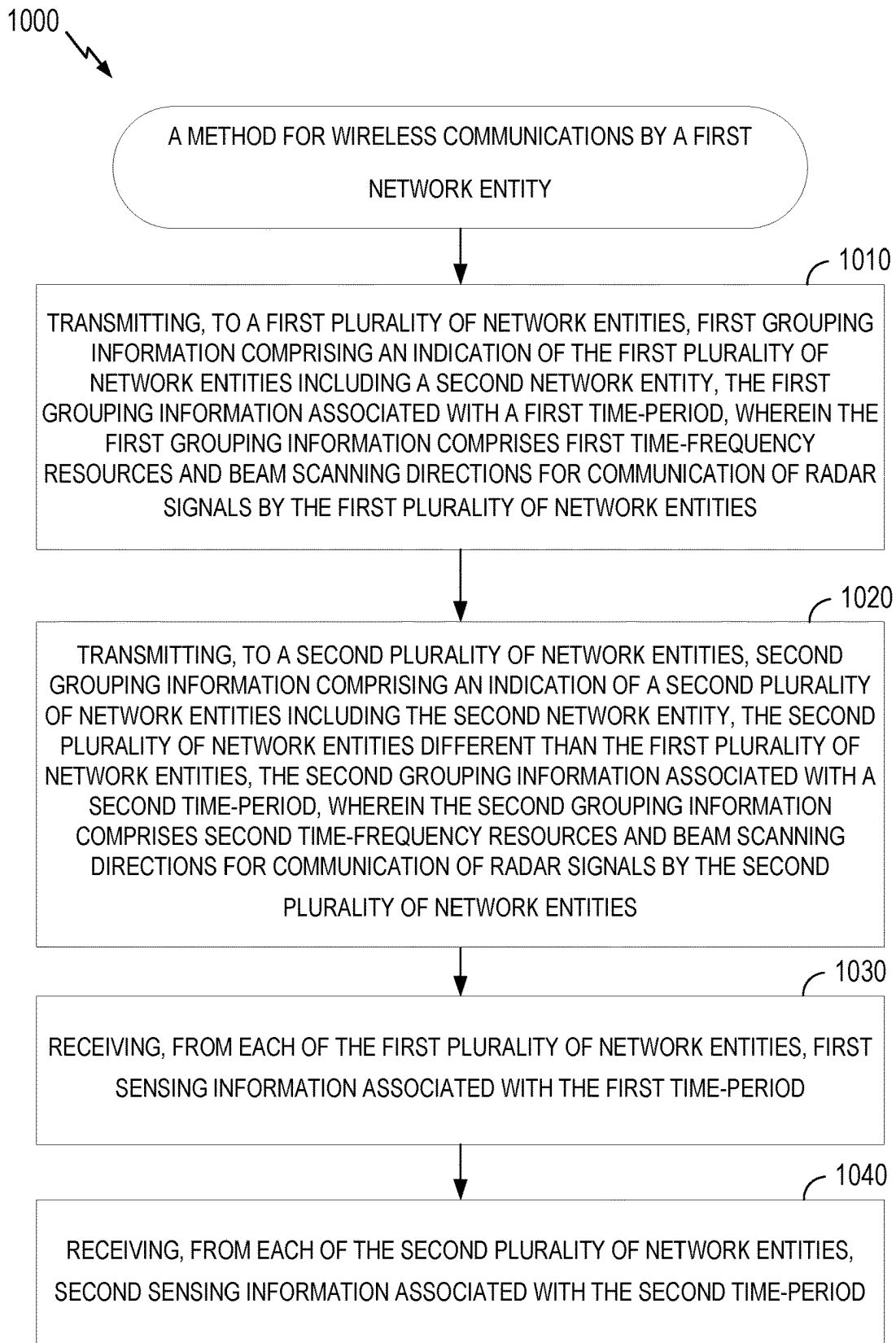


FIG. 7B

**FIG. 8**

**FIG. 9**

**FIG. 10**

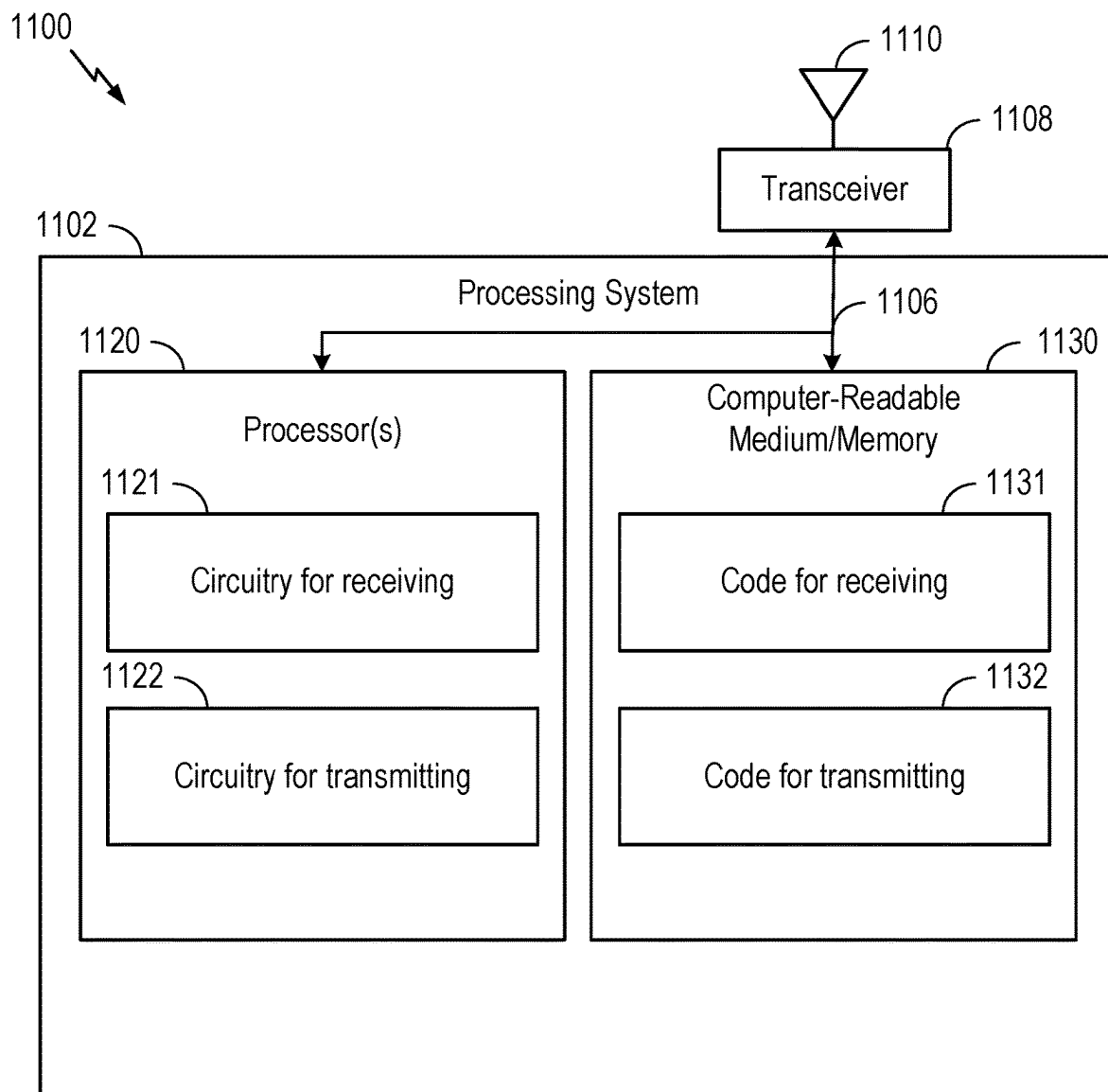
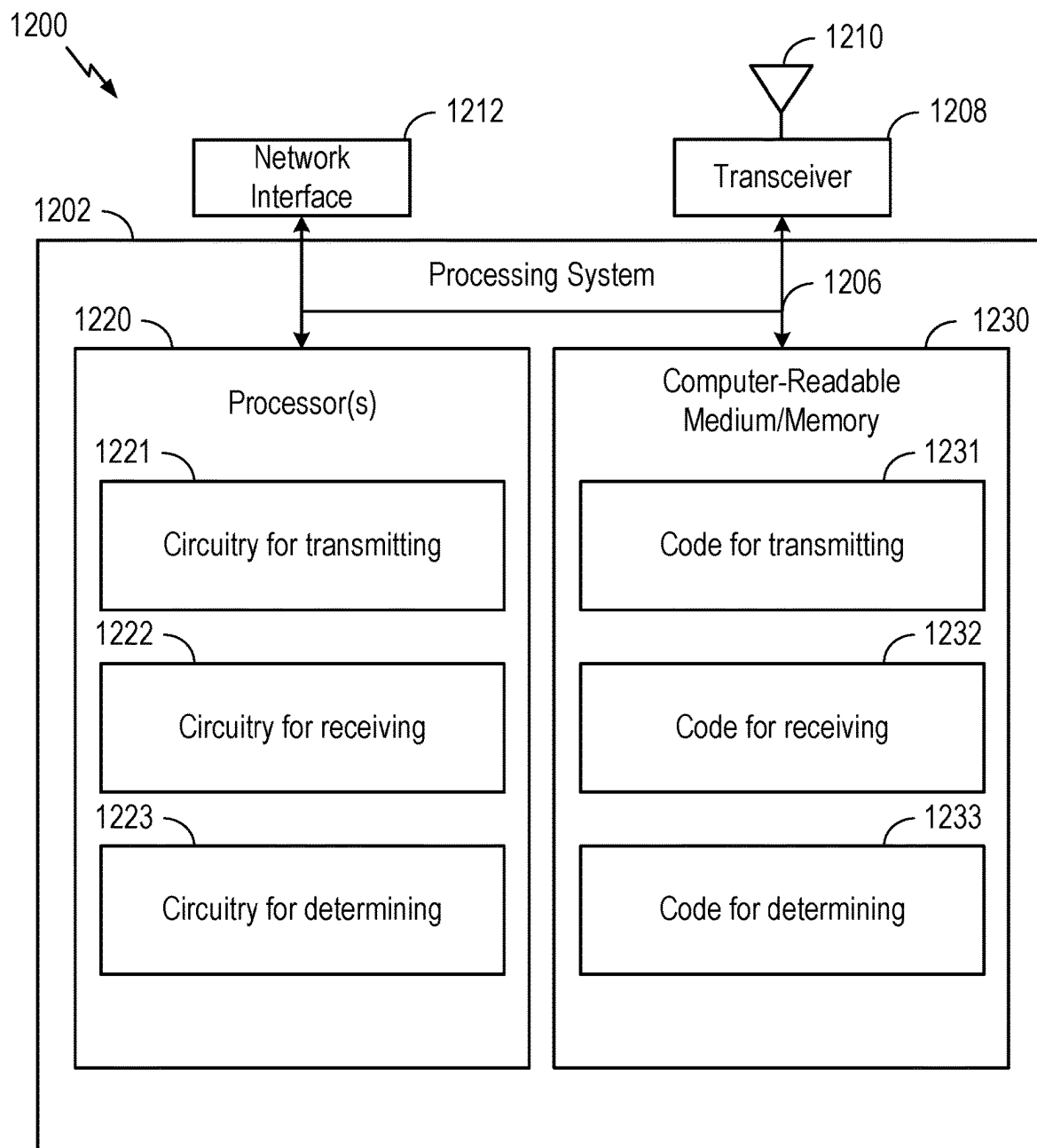


FIG. 11

**FIG. 12**

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ADAPTIVE GROUPING FOR COOPERATIVE RADAR SENSING

INTRODUCTION

Field of the Disclosure

Aspects of the present disclosure relate to wireless communications, and more particularly, to techniques for radar sensing.

Description of Related Art

Wireless communications systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, broadcasts, or other similar types of services. These wireless communications systems may employ multiple-access technologies capable of supporting communications with multiple users by sharing available wireless communications system resources with those users.

Although wireless communications systems have made great technological advancements over many years, challenges still exist. For example, complex and dynamic environments can still attenuate or block signals between wireless transmitters and wireless receivers. Accordingly, there is a continuous desire to improve the technical performance of wireless communications systems, including, for example: improving speed and data carrying capacity of communications, improving efficiency of the use of shared communications mediums, reducing power used by transmitters and receivers while performing communications, improving reliability of wireless communications, avoiding redundant transmissions and/or receptions and related processing, improving the coverage area of wireless communications, increasing the number and types of devices that can access wireless communications systems, increasing the ability for different types of devices to intercommunicate, increasing the number and type of wireless communications mediums available for use, and the like. Consequently, there exists a need for further improvements in wireless communications systems to overcome the aforementioned technical challenges and others.

SUMMARY

One aspect provides a method for wireless communication by a first network entity. The method includes receiving, from a second network entity, first grouping information comprising an indication of a first plurality of network entities including the first network entity, the first grouping information associated with a first time-period. The method includes receiving, from the second network entity, second grouping information comprising an indication of a second plurality of network entities including the first network entity, the second plurality of network entities different than the first plurality of network entities, the second grouping information associated with a second time-period. The method includes receiving a first radar signal during the first time-period and receiving a second radar signal during the second time-period. The method includes transmitting, to the second network entity, first sensing information based on the first radar signal and transmitting, to the second network entity, second sensing information based on the second radar signal.

Another aspect provides another method for wireless communication by a first network entity. The method

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includes transmitting, to a first plurality of network entities, first grouping information comprising an indication of the first plurality of network entities including a second network entity, the first grouping information associated with a first time-period, wherein the first grouping information comprises first time-frequency resources and beam scanning directions for communication of radar signals by the first plurality of network entities. The method includes transmitting, to a second plurality of network entities, second grouping information comprising an indication of a second plurality of network entities including the second network entity, the second plurality of network entities different than the first plurality of network entities, the second grouping information associated with a second time-period, wherein the second grouping information comprises second time-frequency resources and beam scanning directions for communication of radar signals by the second plurality of network entities. The method includes receiving, from each of the first plurality of network entities, first sensing information associated with the first time-period and receiving, from each of the second plurality of network entities, second sensing information associated with the second time-period.

Other aspects provide: an apparatus operable, configured, or otherwise adapted to perform the aforementioned methods as well as those described elsewhere herein; a non-transitory, computer-readable media comprising instructions that, when executed by a processor of an apparatus, cause the apparatus to perform the aforementioned methods as well as those described elsewhere herein; a computer program product embodied on a computer-readable storage medium comprising code for performing the aforementioned methods as well as those described elsewhere herein; and an apparatus comprising means for performing the aforementioned methods as well as those described elsewhere herein. By way of example, an apparatus may comprise a processing system, a device with a processing system, or processing systems cooperating over one or more networks.

The following description and the appended figures set forth certain features for purposes of illustration.

BRIEF DESCRIPTION OF DRAWINGS

The appended figures depict certain features of the various aspects described herein and are not to be considered limiting of the scope of this disclosure.

FIG. 1 depicts an example wireless communications network.

FIG. 2 depicts an example disaggregated base station architecture.

FIG. 3 depicts aspects of an example base station and an example user equipment.

FIGS. 4A, 4B, 4C, and 4D depict various example aspects of data structures for a wireless communications network.

FIG. 5 depicts example cooperative radar sensing.

FIG. 6 depicts a process flow for communications in a network between a radar server and cooperative radar sensing entities.

FIG. 7A depicts an example grouping of cooperative radar sensing entities.

FIG. 7B depicts another example grouping of cooperative radar sensing entities.

FIG. 8 depicts a process flow for communications in a network between a radar server and cooperative radar sensing entities.

FIG. 9 depicts a method for wireless communications.

FIG. 10 depicts a method for wireless communications.

FIG. 11 depicts aspects of an example communications device.

FIG. 12 depicts aspects of an example communications device.

DETAILED DESCRIPTION

Aspects of the present disclosure provide apparatuses, methods, processing systems, and computer-readable mediums for adaptive grouping for cooperative radar sensing.

Radar sensing is a way of estimating distance and/or velocity of objects relative to a radar sensing device. Radar sensing involves the sending of probing signals to targets and inferring useful information from the target echoes (e.g., reflections of the probing signals off the targets). With cooperative radar sensing, two or more devices exchange information and cooperate to perform joint radar sensing over a wider area. Further, cooperative radar sensing may be used to triangulate a position of an object to more accurately determine its position/location.

In cooperative radar sensing, a radar server allocates time, frequency, and/or spatial radio resources to the cooperating radar sensing devices for sending of radar signals. The radar sensing devices transmit radar sensing signals using the allocated radio resources and receive reflections of the radar sensing signals. The radar sensing devices derive sensing information from the transmission and reception of the radar sensing signals, such as the distance and velocity of the targets, and provide the sensing information to the radar server. The radar server collects the information of the cooperating radar sensing devices and generates final cooperative sensing information based on the sensing information collected from the multiple cooperating radar sensing devices.

The number of devices that can cooperate for radar sensing may be limited. For example, the number of cooperating devices may be limited by the amount of available radar signal resources that can be allocated for the radar sensing signals. Further, the number of cooperating devices may be limited by the complexity and overhead for the radar server to coordinate beams and resources for the cooperating radar sensing devices. The number of cooperating devices may be limited by the computation complexity and overhead to generate and report sensing results at each of the cooperating radar sensing devices. With a fixed limited number of cooperating radar sensing devices, there may be holes in the radar sensing coverage.

Accordingly, adaptive grouping of cooperating radar sensing devices is needed. With adaptive grouping of the cooperating radar sensing devices, different coverage areas can be targeted at different times, allowing cooperative radar sensing to be performed over a larger coverage area, even when the number of cooperating radar sensing devices is limited.

Introduction to Wireless Communications Networks

The techniques and methods described herein may be used for various wireless communications networks. While aspects may be described herein using terminology commonly associated with 3G, 4G, and/or 5G wireless technologies, aspects of the present disclosure may likewise be applicable to other communications systems and standards not explicitly mentioned herein.

FIG. 1 depicts an example of a wireless communications network 100, in which aspects described herein may be implemented.

Generally, wireless communications network 100 includes various network entities (alternatively, network

elements or network nodes). A network entity is generally a communications device and/or a communications function performed by a communications device (e.g., a user equipment (UE), a base station (BS), a component of a BS, a server, etc.). For example, various functions of a network as well as various devices associated with and interacting with a network may be considered network entities. Further, wireless communications network 100 includes terrestrial aspects, such as ground-based network entities (e.g., BSs 102), and non-terrestrial aspects, such as satellite 140 and aircraft 145, which may include network entities on-board (e.g., one or more BSs) capable of communicating with other network elements (e.g., terrestrial BSs) and user equipments.

In the depicted example, wireless communications network 100 includes BSs 102, UEs 104, and one or more core networks, such as an Evolved Packet Core (EPC) 160 and 5G Core (5GC) network 190, which interoperate to provide communications services over various communications links, including wired and wireless links.

FIG. 1 depicts various example UEs 104, which may more generally include: a cellular phone, smart phone, session initiation protocol (SIP) phone, laptop, personal digital assistant (PDA), satellite radio, global positioning system, multimedia device, video device, digital audio player, camera, game console, tablet, smart device, wearable device, vehicle, electric meter, gas pump, large or small kitchen appliance, healthcare device, implant, sensor/actuator, display, internet of things (IoT) devices, always on (AON) devices, edge processing devices, or other similar devices. UEs 104 may also be referred to more generally as a mobile device, a wireless device, a wireless communications device, a station, a mobile station, a subscriber station, a mobile subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a remote device, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, and others.

BSs 102 wirelessly communicate with (e.g., transmit signals to or receive signals from) UEs 104 via communications links 120. The communications links 120 between BSs 102 and UEs 104 may include uplink (UL) (also referred to as reverse link) transmissions from a UE 104 to a BS 102 and/or downlink (DL) (also referred to as forward link) transmissions from a BS 102 to a UE 104. The communications links 120 may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity in various aspects.

BSs 102 may generally include: a NodeB, enhanced NodeB (eNB), next generation enhanced NodeB (ng-eNB), next generation NodeB (gNB or gNodeB), access point, base transceiver station, radio base station, radio transceiver, transceiver function, transmission reception point, and/or others. Each of BSs 102 may provide communications coverage for a respective geographic coverage area 110, which may sometimes be referred to as a cell, and which may overlap in some cases (e.g., small cell 102' may have a coverage area 110' that overlaps the coverage area 110 of a macro cell). A BS may, for example, provide communications coverage for a macro cell (covering relatively large geographic area), a pico cell (covering relatively smaller geographic area, such as a sports stadium), a femto cell (relatively smaller geographic area (e.g., a home)), and/or other types of cells.

While BSs 102 are depicted in various aspects as unitary communications devices, BSs 102 may be implemented in various configurations. For example, one or more compo-

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nents of a base station may be disaggregated, including a central unit (CU), one or more distributed units (DUs), one or more radio units (RUs), a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) MC, to name a few examples. In another example, various aspects of a base station may be virtualized. More generally, a base station (e.g., BS 102) may include components that are located at a single physical location or components located at various physical locations. In examples in which a base station includes components that are located at various physical locations, the various components may each perform functions such that, collectively, the various components achieve functionality that is similar to a base station that is located at a single physical location. In some aspects, a base station including components that are located at various physical locations may be referred to as a disaggregated radio access network architecture, such as an Open RAN (O-RAN) or Virtualized RAN (VRAN) architecture. FIG. 2 depicts and describes an example disaggregated base station architecture.

Different BSs 102 within wireless communications network 100 may also be configured to support different radio access technologies, such as 3G, 4G, and/or 5G. For example, BSs 102 configured for 4G LTE (collectively referred to as Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN)) may interface with the EPC 160 through first backhaul links 132 (e.g., an S1 interface). BSs 102 configured for 5G (e.g., 5G NR or Next Generation RAN (NG-RAN)) may interface with 5GC 190 through second backhaul links 184. BSs 102 may communicate directly or indirectly (e.g., through the EPC 160 or 5GC 190) with each other over third backhaul links 134 (e.g., X2 interface), which may be wired or wireless.

Wireless communications network 100 may subdivide the electromagnetic spectrum into various classes, bands, channels, or other features. In some aspects, the subdivision is provided based on wavelength and frequency, where frequency may also be referred to as a carrier, a subcarrier, a frequency channel, a tone, or a subband. For example, 3GPP currently defines Frequency Range 1 (FR1) as including 410 MHz-7125 MHz, which is often referred to (interchangeably) as “Sub-6 GHz”. Similarly, 3GPP currently defines Frequency Range 2 (FR2) as including 24,250 MHz-52,600 MHz, which is sometimes referred to (interchangeably) as a “millimeter wave” (“mmW” or “mmWave”). A base station configured to communicate using mmWave/near mmWave radio frequency bands (e.g., a mmWave base station such as BS 180) may utilize beamforming (e.g., 182) with a UE (e.g., 104) to improve path loss and range.

The communications links 120 between BSs 102 and, for example, UEs 104, may be through one or more carriers, which may have different bandwidths (e.g., 5, 10, 15, 20, 100, 400, and/or other MHz), and which may be aggregated in various aspects. Carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL).

Communications using higher frequency bands may have higher path loss and a shorter range compared to lower frequency communications. Accordingly, certain base stations (e.g., 180 in FIG. 1) may utilize beamforming 182 with a UE 104 to improve path loss and range. For example, BS 180 and the UE 104 may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate the beamforming. In some cases, BS 180 may transmit a beamformed signal to UE 104 in one

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or more transmit directions 182'. UE 104 may receive the beamformed signal from the BS 180 in one or more receive directions 182". UE 104 may also transmit a beamformed signal to the BS 180 in one or more transmit directions 182". BS 180 may also receive the beamformed signal from UE 104 in one or more receive directions 182'. BS 180 and UE 104 may then perform beam training to determine the best receive and transmit directions for each of BS 180 and UE 104. Notably, the transmit and receive directions for BS 180 may or may not be the same. Similarly, the transmit and receive directions for UE 104 may or may not be the same.

Wireless communications network 100 further includes a Wi-Fi AP 150 in communication with Wi-Fi stations (STAs) 152 via communications links 154 in, for example, a 2.4 GHz and/or 5 GHz unlicensed frequency spectrum.

Certain UEs 104 may communicate with each other using device-to-device (D2D) communications link 158. D2D communications link 158 may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), a physical sidelink control channel (PSCCH), and/or a physical sidelink feedback channel (PSFCH).

EPC 160 may include various functional components, including: a Mobility Management Entity (MME) 162, other MMEs 164, a Serving Gateway 166, a Multimedia Broadcast Multicast Service (MBMS) Gateway 168, a Broadcast Multicast Service Center (BM-SC) 170, and/or a Packet Data Network (PDN) Gateway 172, such as in the depicted example. MME 162 may be in communication with a Home Subscriber Server (HSS) 174. MME 162 is the control node that processes the signaling between the UEs 104 and the EPC 160. Generally, MME 162 provides bearer and connection management.

Generally, user Internet protocol (IP) packets are transferred through Serving Gateway 166, which itself is connected to PDN Gateway 172. PDN Gateway 172 provides UE IP address allocation as well as other functions. PDN Gateway 172 and the BM-SC 170 are connected to IP Services 176, which may include, for example, the Internet, an intranet, an IP Multimedia Subsystem (IMS), a Packet Switched (PS) streaming service, and/or other IP services.

BM-SC 170 may provide functions for MBMS user service provisioning and delivery. BM-SC 170 may serve as an entry point for content provider MBMS transmission, may be used to authorize and initiate MBMS Bearer Services within a public land mobile network (PLMN), and/or may be used to schedule MBMS transmissions. MBMS Gateway 168 may be used to distribute MBMS traffic to the BSs 102 belonging to a Multicast Broadcast Single Frequency Network (MBSFN) area broadcasting a particular service, and/or may be responsible for session management (start/stop) and for collecting eMBMS related charging information.

5GC 190 may include various functional components, including: an Access and Mobility Management Function (AMF) 192, other AMFs 193, a Session Management Function (SMF) 194, and a User Plane Function (UPF) 195. AMF 192 may be in communication with Unified Data Management (UDM) 196.

AMF 192 is a control node that processes signaling between UEs 104 and 5GC 190. AMF 192 provides, for example, quality of service (QoS) flow and session management.

Internet protocol (IP) packets are transferred through UPF 195, which is connected to the IP Services 197, and which provides UE IP address allocation as well as other functions

for 5GC **190**. IP Services **197** may include, for example, the Internet, an intranet, an IMS, a PS streaming service, and/or other IP services.

In various aspects, a network entity or network node can be implemented as an aggregated base station, as a disaggregated base station, a component of a base station, an integrated access and backhaul (IAB) node, a relay node, a sidelink node, to name a few examples.

FIG. 2 depicts an example disaggregated base station **200** architecture. The disaggregated base station **200** architecture may include one or more central units (CUs) **210** that can communicate directly with a core network **220** via a backhaul link, or indirectly with the core network **220** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (MC) **225** via an E2 link, or a Non-Real Time (Non-RT) RIC **215** associated with a Service Management and Orchestration (SMO) Framework **205**, or both). A CU **210** may communicate with one or more distributed units (DUs) **230** via respective midhaul links, such as an F1 interface. The DUs **230** may communicate with one or more radio units (RUs) **240** via respective fronthaul links. The RUs **240** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **240**.

Each of the units, e.g., the CUs **210**, the DUs **230**, the RUs **240**, as well as the Near-RT RICs **225**, the Non-RT RICs **215** and the SMO Framework **205**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communications interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally or alternatively, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as a radio frequency (RF) transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

In some aspects, the CU **210** may host one or higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **210**. The CU **210** may be configured to handle user plane functionality (e.g., Central Unit—User Plane (CU-UP)), control plane functionality (e.g., Central Unit—Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **210** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU **210** can be implemented to communicate with the DU **230**, as necessary, for network control and signaling.

The DU **230** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **240**. In some aspects, the DU **230** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling,

modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3rd Generation Partnership Project (3GPP). In some aspects, the DU **230** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **230**, or with the control functions hosted by the CU **210**.

Lower-layer functionality can be implemented by one or more RUs **240**. In some deployments, an RU **240**, controlled by a DU **230**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **240** can be implemented to handle over the air (OTA) communications with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communications with the RU(s) **240** can be controlled by the corresponding DU **230**. In some scenarios, this configuration can enable the DU(s) **230** and the CU **210** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

The SMO Framework **205** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **205** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **205** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **290**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **210**, DUs **230**, RUs **240** and Near-RT RICs **225**. In some implementations, the SMO Framework **205** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **211**, via an O1 interface. Additionally, in some implementations, the SMO Framework **205** can communicate directly with one or more RUs **240** via an O1 interface. The SMO Framework **205** also may include a Non-RT RIC **215** configured to support functionality of the SMO Framework **205**.

The Non-RT RIC **215** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **225**. The Non-RT RIC **215** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **225**. The Near-RT RIC **225** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **210**, one or more DUs **230**, or both, as well as an O-eNB, with the Near-RT RIC **225**.

In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **225**, the Non-RT RIC **215** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **225** and may be received at the SMO Frame-

work **205** or the Non-RT RIC **215** from non-network data sources or from network functions. In some examples, the Non-RT RIC **215** or the Near-RT RIC **225** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **215** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **205** (such as reconfiguration via **01**) or via creation of RAN management policies (such as AI policies).

FIG. 3 depicts aspects of an example BS **102** and a UE **104**.

Generally, BS **102** includes various processors (e.g., **320**, **330**, **338**, and **340**), antennas **334a-t** (collectively **334**), transceivers **332a-t** (collectively **332**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., data source **312**) and wireless reception of data (e.g., data sink **339**). For example, BS **102** may send and receive data between BS **102** and UE **104**. BS **102** includes controller/processor **340**, which may be configured to implement various functions described herein related to wireless communications.

Generally, UE **104** includes various processors (e.g., **358**, **364**, **366**, and **380**), antennas **352a-r** (collectively **352**), transceivers **354a-r** (collectively **354**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., retrieved from data source **362**) and wireless reception of data (e.g., provided to data sink **360**). UE **104** includes controller/processor **380**, which may be configured to implement various functions described herein related to wireless communications.

In regards to an example downlink transmission, BS **102** includes a transmit processor **320** that may receive data from a data source **312** and control information from a controller/processor **340**. The control information may be for the physical broadcast channel (PBCH), physical control format indicator channel (PCFICH), physical HARQ indicator channel (PHICH), physical downlink control channel (PDCCH), group common PDCCH (GC PDCCH), and/or others. The data may be for the physical downlink shared channel (PDSCH), in some examples.

Transmit processor **320** may process (e.g., encode and symbol map) the data and control information to obtain data symbols and control symbols, respectively. Transmit processor **320** may also generate reference symbols, such as for the primary synchronization signal (PSS), secondary synchronization signal (SSS), PBCH demodulation reference signal (DMRS), and channel state information reference signal (CSI-RS).

Transmit (TX) multiple-input multiple-output (MIMO) processor **330** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, and/or the reference symbols, if applicable, and may provide output symbol streams to the modulators (MODs) in transceivers **332a-332t**. Each modulator in transceivers **332a-332t** may process a respective output symbol stream to obtain an output sample stream. Each modulator may further process (e.g., convert to analog, amplify, filter, and upconvert) the output sample stream to obtain a downlink signal. Downlink signals from the modulators in transceivers **332a-332t** may be transmitted via the antennas **334a-334t**, respectively.

In order to receive the downlink transmission, UE **104** includes antennas **352a-352r** that may receive the downlink signals from the BS **102** and may provide received signals to the demodulators (DEMODOs) in transceivers **354a-354r**, respectively. Each demodulator in transceivers **354a-354r** may condition (e.g., filter, amplify, downconvert, and digi-

tize) a respective received signal to obtain input samples. Each demodulator may further process the input samples to obtain received symbols.

MIMO detector **356** may obtain received symbols from all the demodulators in transceivers **354a-354r**, perform MIMO detection on the received symbols if applicable, and provide detected symbols. Receive processor **358** may process (e.g., demodulate, deinterleave, and decode) the detected symbols, provide decoded data for the UE **104** to a data sink **360**, and provide decoded control information to a controller/processor **380**.

In regards to an example uplink transmission, UE **104** further includes a transmit processor **364** that may receive and process data (e.g., for the PUSCH) from a data source **362** and control information (e.g., for the physical uplink control channel (PUCCH)) from the controller/processor **380**. Transmit processor **364** may also generate reference symbols for a reference signal (e.g., for the sounding reference signal (SRS)). The symbols from the transmit processor **364** may be precoded by a TX MIMO processor **366** if applicable, further processed by the modulators in transceivers **354a-354r** (e.g., for SC-FDM), and transmitted to BS **102**.

At BS **102**, the uplink signals from UE **104** may be received by antennas **334a-t**, processed by the demodulators in transceivers **332a-332t**, detected by a MIMO detector **336** if applicable, and further processed by a receive processor **338** to obtain decoded data and control information sent by UE **104**. Receive processor **338** may provide the decoded data to a data sink **339** and the decoded control information to the controller/processor **340**.

Memories **342** and **382** may store data and program codes for BS **102** and UE **104**, respectively.

Scheduler **344** may schedule UEs for data transmission on the downlink and/or uplink.

In various aspects, BS **102** may be described as transmitting and receiving various types of data associated with the methods described herein. In these contexts, “transmitting” may refer to various mechanisms of outputting data, such as outputting data from data source **312**, scheduler **344**, memory **342**, transmit processor **320**, controller/processor **340**, TX MIMO processor **330**, transceivers **332a-t**, antenna **334a-t**, and/or other aspects described herein. Similarly, “receiving” may refer to various mechanisms of obtaining data, such as obtaining data from antennas **334a-t**, transceivers **332a-t**, RX MIMO detector **336**, controller/processor **340**, receive processor **338**, scheduler **344**, memory **342**, and/or other aspects described herein.

In various aspects, UE **104** may likewise be described as transmitting and receiving various types of data associated with the methods described herein. In these contexts, “transmitting” may refer to various mechanisms of outputting data, such as outputting data from data source **362**, memory **382**, transmit processor **364**, controller/processor **380**, TX MIMO processor **366**, transceivers **354a-t**, antenna **352a-t**, and/or other aspects described herein. Similarly, “receiving” may refer to various mechanisms of obtaining data, such as obtaining data from antennas **352a-t**, transceivers **354a-t**, RX MIMO detector **356**, controller/processor **380**, receive processor **358**, memory **382**, and/or other aspects described herein.

In some aspects, a processor may be configured to perform various operations, such as those associated with the methods described herein, and transmit (output) to or receive (obtain) data from another interface that is configured to transmit or receive, respectively, the data.

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FIGS. 4A, 4B, 4C, and 4D depict aspects of data structures for a wireless communications network, such as wireless communications network 100 of FIG. 1.

In particular, FIG. 4A is a diagram 400 illustrating an example of a first subframe within a 5G (e.g., 5G NR) frame structure, FIG. 4B is a diagram 430 illustrating an example of DL channels within a 5G subframe, FIG. 4C is a diagram 450 illustrating an example of a second subframe within a 5G frame structure, and FIG. 4D is a diagram 480 illustrating an example of UL channels within a 5G subframe.

Wireless communications systems may utilize orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) on the uplink and downlink. Such systems may also support half-duplex operation using time division duplexing (TDD). OFDM and single-carrier frequency division multiplexing (SC-FDM) partition the system bandwidth (e.g., as depicted in FIGS. 4B and 4D) into multiple orthogonal subcarriers. Each subcarrier may be modulated with data. Modulation symbols may be sent in the frequency domain with OFDM and/or in the time domain with SC-FDM.

A wireless communications frame structure may be frequency division duplex (FDD), in which, for a particular set of subcarriers, subframes within the set of subcarriers are dedicated for either DL or UL. Wireless communications frame structures may also be time division duplex (TDD), in which, for a particular set of subcarriers, subframes within the set of subcarriers are dedicated for both DL and UL.

In FIGS. 4A and 4C, the wireless communications frame structure is TDD where D is DL, U is UL, and X is flexible for use between DL/UL. UEs may be configured with a slot format through a received slot format indicator (SFI) (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling). In the depicted examples, a 10 ms frame is divided into 10 equally sized 1 ms subframes. Each subframe may include one or more time slots. In some examples, each slot may include 7 or 14 symbols, depending on the slot format. Subframes may also include mini-slots, which generally have fewer symbols than an entire slot. Other wireless communications technologies may have a different frame structure and/or different channels.

In certain aspects, the number of slots within a subframe is based on a slot configuration and a numerology. For example, for slot configuration 0, different numerologies (μ) 0 to 5 allow for 1, 2, 4, 8, 16, and 32 slots, respectively, per subframe. For slot configuration 1, different numerologies 0 to 2 allow for 2, 4, and 8 slots, respectively, per subframe. Accordingly, for slot configuration 0 and numerology μ , there are 14 symbols/slot and 2μ slots/subframe. The subcarrier spacing and symbol length/duration are a function of the numerology. The subcarrier spacing may be equal to $2^\mu \times 15$ kHz, where μ is the numerology 0 to 5. As such, the numerology $\mu=0$ has a subcarrier spacing of 15 kHz and the numerology $\mu=5$ has a subcarrier spacing of 480 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. 4A, 4B, 4C, and 4D provide an example of slot configuration 0 with 14 symbols per slot and numerology $\mu=2$ with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μ s.

As depicted in FIGS. 4A, 4B, 4C, and 4D, a resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends, for example, 12 consecutive subcarriers. The resource grid is divided into multiple

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resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.

As illustrated in FIG. 4A, some of the REs carry reference (pilot) signals (RS) for a UE (e.g., UE 104 of FIGS. 1 and 3). The RS may include demodulation RS (DMRS) and/or channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and/or phase tracking RS (PT-RS).

FIG. 4B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs), each CCE including, for example, nine RE groups (REGs), each REG including, for example, four consecutive REs in an OFDM symbol.

A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE (e.g., 104 of FIGS. 1 and 3) to determine subframe/symbol timing and a physical layer identity.

A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing.

Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the aforementioned DMRS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block. The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and/or paging messages.

As illustrated in FIG. 4C, some of the REs carry DMRS (indicated as R for one particular configuration, but other DMRS configurations are possible) for channel estimation at the base station. The UE may transmit DMRS for the PUCCH and DMRS for the PUSCH. The PUSCH DMRS may be transmitted, for example, in the first one or two symbols of the PUSCH. The PUCCH DMRS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. UE 104 may transmit sounding reference signals (SRS). The SRS may be transmitted, for example, in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.

FIG. 4D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and HARQ ACK/NACK feedback. The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

Aspects Related to Adaptive Cooperative Radar Sensing

As discussed above, radar sensing is way of estimating distance and/or velocity of objects and with cooperative radar sensing, two or more devices exchange information and jointly perform radar sensing. FIG. 5 depicts example cooperative radar sensing. In the example shown in FIG. 5, radar sensing entity 504, radar sensing entity 506, radar

sensing entity **508**, and radar server **510** are configured to perform cooperative radar sensing of a target **502**. Target **502** may be an example of a UE **104** in wireless communications network **100** illustrated in FIGS. 1-3. Radar sensing entities **504**, **506**, and **508** may be network entities. In some examples, radar sensing entities **504**, **506**, and **508** are an example of BSs **102** in wireless communications network **100** illustrated in FIGS. 1-3. In some examples, a radar server may also be one of the cooperative radar sensing entities.

Although BSs are shown in FIG. 5, a UE (e.g., such as UE **104** in wireless communications network **100** illustrated in FIGS. 1-3) may also be a cooperating radar sensing entity. Further, it should be understood that while FIG. 5 depicts three entities performing cooperative radar sensing, this is merely one illustrative example, and fewer than three or more than three entities may perform cooperative radar sensing. Further, it should be understood that while FIG. 5 depicts cooperative radar sensing of one target, this is merely one illustrative example, and cooperative radar sensing may be performed for more than one target.

FIG. 6 depicts a process flow **600** for communications in a network between radar server **510** and the cooperating radar sensing entities. Process flow **600** depicts the communications between radar server **510**, radar sensing entity **504**, and radar sensing entity **506**. It is noted that radar sensing entity **508** is not shown in FIG. 6 for simplicity, however, it should be understood that similar communications are performed with radar sensing entity **508** as those shown for radar sensing entity **504** and **506**.

In order to allocate radio resources for the radar signaling, radar server **510** has coverage and beam information of each of the cooperating radar sensing entities, radar sensing entities **504** and **506**. The coverage information may include the transmit power and/or the effective isotropic radiated power (EIRP) available at the cooperating radar sensing devices. The beam information may include the number of beams and/or the beam directions available at the cooperating radar sensing devices.

In some examples, the coverage and/or beam information is stored at the radar server **510** when the radar server **510** is deployed. In some examples, the coverage and/or beam information is signaled to the radar server **510** by the respective cooperating radar sensing entities **504** and **506**, at **602** and **604**, respectively, as shown in FIG. 6. In some examples, a portion of the beam and/or coverage information is stored at the radar server **510** when the radar server **510** is deployed and a portion of the beam and/or coverage information is signaled to the radar server **510** by the cooperating radar sensing entities **504** and **506**. The beam and/or coverage information may be signaled by the cooperating radar sensing entities **504** and **506** to radar server **510** periodically or on-demand.

It is noted that while FIG. 6 depicts the coverage and/or beam information from radar sensing entity **506** being signaled at **604** after the coverage and/or beam information from radar sensing entity **504** at **602**, it should be understood that the coverage and/or beam information from radar sensing entity **506** at **604** may be signaled at the same time or before the coverage and/or beam information from radar sensing entity **504** at **602**.

Radar server **510** uses the beam and/or coverage information to determine time, frequency, and/or spatial radio resources for the cooperating radar sensing entities **504** and **506**. The spatial resources may include beam scanning directions for radar signal transmission.

Radar server **510** provides a radar signal configuration to the cooperating radar sensing entities to provide the respective radio resource allocations to the respective radar sensing entities to use for radar signaling. As shown in FIG. 6, radar server **510** provides a radar signal configuration to radar sensing entity **504** at **606** and provides a radar signal configuration to radar sensing entity **506** at **608**. The radio resource allocations may also be used by the radar sensing entities for communication.

Radar sensing parameters may include a range resolution (Δd), a maximum range (d_{max}), a velocity resolution (Δv), a maximum velocity (v_{max}), and an effective velocity range ($v_{eff}=2v_{max}$). In some examples, radar signal parameters are configured. The radar signal parameters may include a subcarrier spacing (Δf). The minimum SCS may be about ten times a maximum Doppler frequency, where $\Delta f=20 v_{max} f_c/c$ giving $v_{max}<c \Delta f/20 f_c$. The radar signal parameters may include a guard interval or cyclic prefix length (T_g). The guard interval or CP is for a time interval until the latest radar echo is received, $T_g>d_{max}/c$ giving $d_{max}<c T_g/2$. The radar signal parameters may include a bandwidth (W) of the radar signal, where $W>c/(2\Delta d)$ giving $\Delta d>c/(2 W)$. The radar signal parameters may include a burst duration (T_B) of the radar signal, where $T_B>c/(2 f_c \Delta v)$ giving $\Delta v>c/(2 f_c T_B)$. The radar signal parameters may include a time domain spacing (T_s) between consecutive radar signals, where $T_s<c/(2 f_c v_{eff})$ giving $v_{eff}<c/(2 f_c T_s)$. The radar signal parameters may include a frequency domain spacing (f_s) between consecutive radar signals, where $f_s<c/(2 d_{max})$ giving $d_{max}<c/(2 f_s)$.

It is noted that while FIG. 6 depicts the radar signal configuration being signaled to radar sensing entity **506** at **608** after the radar signal configuration to radar sensing entity **504** at **606**, it should be understood that the radar signal configuration signaled to radar sensing entity **506** at **608** may be signaled at the same time or before the radar signal configuration to radar sensing entity **504** at **606**.

Radar server **510** may also provide the radar sensing entities **504** and **506** with the radar signal configurations of neighboring radar sensing entities. For example, radar server **510** may provide each of the radar sensing entities **504**, **506**, and **508** with the radar signal configurations of each of the other radar sensing entities **504**, **506**, and **508**. As shown in FIG. 6, radar server **510** provides the radar signal configurations of the neighbor sensing entities to radar sensing entity **504** at **610** and provides the radar signal configurations of neighbor sensing entities to radar sensing entity **506** at **612**. The radio signal configurations of the neighbor sensing entities may be useful in cases where the radar sensing entities are configured for bistatic radar or multi-static radar. In bistatic and multi-static radar, an entity that receives a reflected radar signal is a different entity than the entity that transmitted the radar signal. In monostatic radar, the transmitter and receiver are colocated.

It is noted that while FIG. 6 depicts the radar signal configurations of neighboring sensing entities being signaled to radar sensing entity **506** at **612** after the radar signal configurations of neighboring sensor entities to radar sensing entity **504** at **610**, it should be understood that the radar signal configurations of neighboring sensor entities may be signaled to radar sensing entity **506** at **612** at the same time or before the radar signal configurations of neighboring sensor entities to radar sensing entity **504** at **610**. Further, while the radar sensing configurations for the cooperating radar sensing entities **504** and **506** are shown as signaled to the cooperating radar sensing entities separately from the radar sensing configurations of neighboring sensor entities,

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the radar sensing configurations of neighboring sensor entities at 610 and 612 may be signaled together with the radar sensing configurations for the cooperating radar sensing entities 504 and 506, at 606 and 608, respectively.

The cooperative radar sensing entities can then transmit radar sensing signals using the allocations received from radar server 510. As shown in FIG. 6, radar sensing entity 504 transmits one or more radar signal(s) at 614 and radar sensing entity 506 transmits one or more radar signal(s) at 616. In some examples, the radar sensing signals are radar reference signals (RSSs). The radar signals may be directed at target 502. Although not shown, the radar sensing entities may also perform communications simultaneously with the radar sensing.

It is noted that while FIG. 6 depicts the radar signals from radar sensing entity 506 at 616 being transmitted after the radar signals from radar sensing entity 504 at 614, it should be understood that the radar signals from radar sensing entity 506 at 616 may be transmitted at the same time or before the radar signals from radar sensing entity 504 at 614.

The radar sensing entity 504 and 506 monitor and receive reflections of the radar sensing signals. As discussed above, in bistatic or multi-static radar, radar sensing entity 506 may receive a radar signal transmitted by radar sensing entity 504 that was reflected by target 502. Similarly, radar sensing entity 504 may receive a radar signal transmitted by radar sensing entity 506 that was reflected by target 502. Additionally or alternatively, radar sensing entity 504 may receive a radar signal transmitted by radar sensing entity 504 and reflected by target 502 and radar sensing entity 506 may receive a radar signal transmitted by radar sensing entity 506 and reflected by target 502.

The radar sensing entities 504 and 506 may derive radar sensing information based on the received reflected radar signal(s). For example, radar sensing information may include the distance and/or the velocity of target 502. The radar sensing entities 504 and 506 may derive the distance of target 502 based on a delay between when a radar signal was transmitted and when a reflection of the radar signal was received. The radar sensing entities 504 and 506 may derive the velocity of target 502 based on Doppler information of the received reflected radar signals from target 502. Additionally or alternatively, radar sensing entities 504 and 506 may provide the delay and/or Doppler information to radar server 510.

As shown in FIG. 6, radar sensing entity 504 transmits radar sensing information to radar server 510 at 618 and radar sensing entity 506 transmits radar sensing information to radar server 510 at 620. The radar sensing information may be transmitted to radar server 510 in a radar sensing report.

It is noted that while FIG. 6 depicts the radar sensing information from radar sensing entity 506 being transmitted at 620 after the radar sensing information from radar sensing entity 504 at 618, it should be understood that the radar sensing information from radar sensing entity 506 at 620 may be transmitted at the same time or before the radar sensing information from radar sensing entity 504 at 618.

Radar server 510 collects the radar sensing information of the cooperating radar sensing devices and generates final cooperative sensing information based on the sensing information collected from the multiple cooperating radar sensing devices. In some examples, the final cooperative sensing information may be used in managing interference in the system. In some examples, the final cooperative sensing information may be used for coordinating the beams used by the radar sensing entities.

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According to certain aspects, the cooperating radar sensing entities can be adaptively grouped. For example, a large set of radar sensing entities can be divided into smaller groups, where the radar sensing entities within each group perform cooperative radar sensing with each other. The groups may each communicate with the same radar server.

FIG. 7A depicts an example grouping 700A of cooperative radar sensing entities for radar sensing of one or more targets 716, 718, and/or 720, which may be example UEs 104 in wireless communications network 100 in FIG. 1. As shown, the radar sensing entities 504, 506, and 508 may be in a network with additional radar sensing entities 710, 712, and 714. As shown, radar sensing entities 504, 506, and 508 are configured in a radar sensing group 722 and radar sensing entities 710, 712, and 714 are configured in another radar sensing group 724. The grouping 700A may be for cooperative radar sensing during a first period.

The radar sensing entities may be adaptively regrouped for cooperative radar sensing during a second period. According to certain aspects, the groups may be formed based on target coverage areas, an availability of radio resources, and/or a capability of the radar sensing entities. In some examples, the radar server can determine the adaptive groupings based on the sensing information received from the radar sensing entities (e.g., at 618 and 620). In some examples, the adaptive regrouping may be done to track a moving target. In some examples, the sensing information can be used to determine the direction a target is moving (e.g., in addition to the distance and velocity) and can adaptively (re)group radar sensing entities in the direction of the target's motion.

Additionally or alternatively, the radar server can determine the adaptive groupings based on the coverage and/or beam information received from the radar sensing entities (e.g., at 602 and 604). For example, the radar server can use the coverage and/or beam information to determine coverage holes and can adaptively (re)group radar sensing entities in order to fill the coverage holes. For example, grouping 700A alone, may lead to a hole in radar coverage at the position of target 718. However, by further using grouping 700B at a different time, radar coverage is provided by radar sensing group 728 at the position of target 718.

FIG. 7B depicts another example grouping 700B of cooperating radar sensing entities. As shown, radar sensing entities 508, 710, and 712 are adaptively regrouped into a radar sensing group 728 for cooperative sensing during the second period. Radar sensing entities 504 and 506 may be adaptively regrouped into radar sensing group 726 for cooperative sensing during the second period and radar sensing entity 714 may be adaptively regrouped into yet another radar sensing group 730 for cooperative sensing during the second period.

According to certain aspects, the radar server signals the adaptive grouping to the radar sensing entities. In some examples, the radar server includes radar sensing entity grouping information in the radar signal configuration as shown in the call flow 800 at 806 and 808 in FIG. 8. For example, radar server 510 may indicate the radar sensing group 722 to radar sensing entities 504 and 506 in the radar signal configurations at 806 and 808 respectively. In some examples, the radar server signals the radar sensing entity grouping information separately from the radar signal configuration. The radar signal configuration and/or the radar sensing entity grouping information may be unicast, broadcast, or multicast to the radar sensing entities 504 and 506.

According to certain aspects, the radar sensing grouping information includes the identifier(s) of the radar sensing

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entities (e.g., a UE ID or a gNB ID) associated with a group and the radio resource information for that radar sensing group. In some examples, the sequence (e.g., the value of the sequence of reference symbols of the radar signal) of the radar signals is based on the identifier(s) of the radar sensing entities. Thus, the radar sensing entities in a radar sensing group are provided with the identifier(s) of the other radar sensing entities in the radar sensing group so that the radar sensing entities can identify the sender of a received radar echo based on the sequence.

In some examples, the identifier(s) of the radar sensing entities can be used to determine the radio resources for radar signaling. In some examples, the grouping information only includes identifier(s) of the radar sensing entities for the radar sensing group. In this case, a mapping of groupings to radio resources and beams may be previously configured. Thus, when a radar sensing entity is indicated a radar sensing group, the radar sensing entity can determine the radio resources and beams associated with the indicated radar sensing group based on the mapping.

In some examples, the grouping information configures different groupings of radar sensing entities for different time-periods. In some examples, the different groupings and the different time-periods are configured in the radar signal configuration. For example, radar server 510 may indicate the radar sensing group 722 and the associated first time-period and the radar sensing group 726 and the associated second time-period to radar sensing entities 504 and 506 in the radar signal configurations at 806 and 808 respectively. Additionally or alternatively, the different groupings of radar sensing entities for the different time-periods may be configured in different radar signal configurations at different times. In some examples, the radar server may specify a time-period in which a grouping of cooperative radar sensing entities is configured by indicating a time offset and time duration. The radar server may indicate the time offset and time duration in terms of system frame numbers and slot numbers.

Example Operations of a Radar Sensing Entity

FIG. 9 shows a method 900 for wireless communications by a first radar sensing entity, which may be a first network entity such as a BS 102 or a UE 104 of FIGS. 1 and 3.

Method 900 begins at 910 with receiving, from a second network entity, first grouping information comprising an indication of a first plurality of network entities including the first network entity, the first grouping information associated with a first time-period.

Method 900 then proceeds to step 920 with receiving, from the second network entity, second grouping information comprising an indication of a second plurality of network entities including the first network entity, the second plurality of network entities different than the first plurality of network entities, the second grouping information associated with a second time-period.

Method 900 then proceeds to step 930 with receiving a first radar signal during the first time-period.

Method 900 then proceeds to step 940 with receiving a second radar signal during the second time-period.

Method 900 then proceeds to step 950 with transmitting, to the second network entity, first sensing information based on the first radar signal.

Method 900 then proceeds to step 960 with transmitting, to the second network entity, second sensing information based on the second radar signal.

In one aspect, the first grouping information comprises a network entity identifier for each of the first plurality of network entities.

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In one aspect, the first grouping information comprises an indication of the first time-period.

In one aspect, the first grouping information comprises time-frequency resources for communication of radar signals by the first plurality of network entities.

In one aspect, the first grouping information further comprises beam scanning directions for communication of radar signals by the first plurality of network entities.

In one aspect, method 900 further include transmitting, during the first time-period, a third radar signal according to the first grouping information.

In one aspect, method 900 further include transmitting, during the first time-period, the first radar signal according to the first grouping information.

In one aspect, the first sensing information comprises at least one of range or velocity information of a target object.

In one aspect, method 900 further include transmitting, to the second network entity, one or more of: a transmit power of the first network entity, an effective isotropic radiated power of the first network entity, a number of beams of the first network entity, or direction information for each of the beams of the first network entity.

In one aspect, method 900, or any aspect related to it, may be performed by an apparatus, such as communications device 1100 of FIG. 11, which includes various components operable, configured, or adapted to perform the method 900. Communications device 1100 is described below in further detail.

Note that FIG. 9 is just one example of a method, and other methods including fewer, additional, or alternative steps are possible consistent with this disclosure.

Example Operations of a Radar Server

FIG. 10 shows a method 1000 for wireless communications by a radar server, which may be a first network entity, such as BS 102 of FIGS. 1 and 3, or a disaggregated base station as discussed with respect to FIG. 2, or may be a different network entity in wireless communications network 100 illustrates in FIG. 1.

Method 1000 begins at 1010 with transmitting, to a first plurality of network entities, first grouping information comprising an indication of the first plurality of network entities including a second network entity, the first grouping information associated with a first time-period, wherein the first grouping information comprises first time-frequency resources and beam scanning directions for communication of radar signals by the first plurality of network entities;

Method 1000 then proceeds to step 1020 with transmitting, to a second plurality of network entities, second grouping information comprising an indication of a second plurality of network entities including the second network entity, the second plurality of network entities different than the first plurality of network entities, the second grouping information associated with a second time-period, wherein the second grouping information comprises second time-frequency resources and beam scanning directions for communication of radar signals by the second plurality of network entities;

Method 1000 then proceeds to step 1030 with receiving, from each of the first plurality of network entities, first sensing information associated with the first time-period; and

Method 1000 then proceeds to step 1040 with receiving, from each of the second plurality of network entities, second sensing information associated with the second time-period.

In one aspect, the first grouping information comprises a network entity identifier for each of the first plurality of network entities.

In one aspect, the first grouping information comprises an indication of the first time-period.

In one aspect, the first sensing information comprises at least one of range or velocity information of a target object.

In one aspect, method **1000** further includes receiving, from each of the first plurality of network entities and the second plurality of network entities, one or more of: a transmit power of the network entity, an effective isotropic radiated power of the network entity, a number of beams of the network entity, or direction information for each of the beams of the network entity.

In one aspect, method **1000** further includes determining the second grouping information based on the one or more of: the transmit power of the network entity, the effective isotropic radiated power of the network entity, the number of beams of the network entity, or the direction information for each of the beams of the network entity.

In one aspect, method **1000** further includes determining the second grouping information based on the first sensing information.

In one aspect, method **1000**, or any aspect related to it, may be performed by an apparatus, such as communications device **1200** of FIG. **12**, which includes various components operable, configured, or adapted to perform the method **1000**. Communications device **1200** is described below in further detail.

Note that FIG. **10** is just one example of a method, and other methods including fewer, additional, or alternative steps are possible consistent with this disclosure.

Example Communications Devices

FIG. **11** depicts aspects of an example communications device **1100**. In some aspects, communications device **1100** is a radar sensing entity which may be a network entity, such as BS **102** or a UE **104** described above with respect to FIGS. **1** and **3**.

The communications device **1100** includes a processing system **1102** coupled to a transceiver **1108** (e.g., a transmitter and/or a receiver). The transceiver **1108** is configured to transmit and receive signals for the communications device **1100** via an antenna **1110**, such as the various signals as described herein. The processing system **1102** may be configured to perform processing functions for the communications device **1100**, including processing signals received and/or to be transmitted by the communications device **1100**.

The processing system **1102** includes one or more processors **1120**. In various aspects, the one or more processors **1120** may be representative of one or more of receive processor **358**, transmit processor **364**, TX MIMO processor **366**, and/or controller/processor **380**, as described with respect to FIG. **3**. The one or more processors **1120** are coupled to a computer-readable medium/memory **1130** via a bus **1106**. In certain aspects, the computer-readable medium/memory **1130** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **1120**, cause the one or more processors **1120** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it. Note that reference to a processor performing a function of communications device **1100** may include one or more processors performing that function of communications device **1100**.

In the depicted example, computer-readable medium/memory **1130** stores code (e.g., executable instructions) **1131** for receiving and code **1132** for receiving. Processing of the code **1131-1132** may cause the communications device **1100** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it.

The one or more processors **1120** include circuitry configured to implement (e.g., execute) the code stored in the computer-readable medium/memory **1130**, including circuitry **1121** for receiving and circuitry **1122** for transmitting.

Processing with circuitry **1121-1122** may cause the communications device **1100** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it.

Various components of the communications device **1100** may provide means for performing the method **900** described with respect to FIG. **9**, or any aspect related to it. For example, means for transmitting, sending or outputting for transmission may include the transceivers **354** and/or antenna(s) **352** of the UE **104** or the transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3**, and/or transceiver **1108** and antenna **1110** of the communications device **1100** in FIG. **11**. Means for receiving or obtaining may include the transceivers **354** and/or antenna(s) **352** of the UE **104** or the transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or transceiver **1108** and antenna **1110** of the communications device **1100** in FIG. **11**.

FIG. **12** depicts aspects of an example communications device. In some aspects, communications device **1200** is a radar sever which may be a network entity, such as a BS **102** of FIGS. **1** and **3**, or a disaggregated base station as discussed with respect to FIG. **2**.

The communications device **1200** includes a processing system **1202** coupled to a transceiver **1208** (e.g., a transmitter and/or a receiver) and/or a network interface **1212**. The transceiver **1208** is configured to transmit and receive signals for the communications device **1200** via an antenna **1210**, such as the various signals as described herein. The network interface **1212** is configured to obtain and send signals for the communications device **1200** via communications link(s), such as a backhaul link, midhaul link, and/or fronthaul link as described herein, such as with respect to FIG. **2**. The processing system **1202** may be configured to perform processing functions for the communications device **1200**, including processing signals received and/or to be transmitted by the communications device **1200**.

The processing system **1202** includes one or more processors **1220**. In various aspects, one or more processors **1220** may be representative of one or more of receive processor **338**, transmit processor **320**, TX MIMO processor **330**, and/or controller/processor **340**, as described with respect to FIG. **3**. The one or more processors **1220** are coupled to a computer-readable medium/memory **1230** via a bus **1206**. In certain aspects, the computer-readable medium/memory **1230** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **1220**, cause the one or more processors **1220** to perform the method **1000** described with respect to FIG. **10**, or any aspect related to it. Note that reference to a processor of communications device **1200** performing a function may include one or more processors of communications device **1200** performing that function.

In the depicted example, the computer-readable medium/memory **1230** stores code (e.g., executable instructions) **1231** for transmitting, code **1232** for receiving, and/or code **1233** for determining. Processing of the code **1231-1233** may cause the communications device **1200** to perform the method **1000** described with respect to FIG. **10**, or any aspect related to it.

The one or more processors **1220** include circuitry configured to implement (e.g., execute) the code stored in the computer-readable medium/memory **1230**, including circuitry **1221** for transmitting, circuitry **1222** for receiving, and/or circuitry **1223** for determining. Processing with cir-

cuitry **1221-1223** may cause the communications device **1200** to perform the method **1000** as described with respect to FIG. **10**, or any aspect related to it.

Various components of the communications device **1200** may provide means for performing the method **1000** as described with respect to FIG. **10**, or any aspect related to it. Means for transmitting, sending or outputting for transmission may include the transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or transceiver **1208** and antenna **1210** of the communications device **1200** in FIG. **12**. Means for receiving or obtaining may include the transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or transceiver **1208** and antenna **1210** of the communications device **1200** in FIG. **12**.

Example Clauses

Implementation examples are described in the following numbered clauses:

Clause 1: A method for wireless communications by a first network entity, the method comprising: receiving, from a second network entity, first grouping information comprising an indication of a first plurality of network entities including the first network entity, the first grouping information associated with a first time-period; receiving, from the second network entity, second grouping information comprising an indication of a second plurality of network entities including the first network entity, the second plurality of network entities different than the first plurality of network entities, the second grouping information associated with a second time-period; receiving a first radar signal during the first time-period; receiving a second radar signal during the second time-period; transmitting, to the second network entity, first sensing information based on the first radar signal; and transmitting, to the second network entity, second sensing information based on the second radar signal.

Clause 2: The method of clause 1, wherein the first grouping information comprises a network entity identifier for each of the first plurality of network entities.

Clause 3: The method of any one or more of clauses 1-2, wherein the first grouping information comprises an indication of the first time-period.

Clause 4: The method of any one or more of clauses 1-3, wherein the first grouping information comprises time-frequency resources for communication of radar signals by the first plurality of network entities.

Clause 5: The method of any one or more of clauses 1-4, wherein the first grouping information further comprises beam scanning directions for communication of radar signals by the first plurality of network entities.

Clause 6: The method of any one or more of clauses 1-5, further comprising transmitting, during the first time-period, a third radar signal according to the first grouping information.

Clause 7: The method of any one or more of clauses 1-6, further comprising transmitting, during the first time-period, the first radar signal according to the first grouping information.

Clause 8: The method of any one or more of clauses 1-7, wherein the first sensing information comprises at least one of range or velocity information of a target object.

Clause 9: The method of any one or more of clauses 1-8, further comprising transmitting, to the second network entity, one or more of: a transmit power of the first network entity, an effective isotropic radiated power of the first network entity, a plurality of beams of the first network

entity, or direction information for each of the plurality of beams of the first network entity.

Clause 10: A method for wireless communications by a first network entity, the method comprising: transmitting, to a first plurality of network entities, first grouping information comprising an indication of the first plurality of network entities including a second network entity, the first grouping information associated with a first time-period, wherein the first grouping information comprises first time-frequency resources and beam scanning directions for communication of radar signals by the first plurality of network entities; transmitting, to a second plurality of network entities, second grouping information comprising an indication of a second plurality of network entities including the second network entity, the second plurality of network entities different than the first plurality of network entities, the second grouping information associated with a second time-period, wherein the second grouping information comprises second time-frequency resources and beam scanning directions for communication of radar signals by the second plurality of network entities; receiving, from each of the first plurality of network entities, first sensing information associated with the first time-period; and receiving, from each of the second plurality of network entities, second sensing information associated with the second time-period.

Clause 11: The method of clause 10, wherein the first grouping information comprises a network entity identifier for each of the first plurality of network entities.

Clause 12: The method of any one or more of clauses 10-11, wherein the first grouping information comprises an indication of the first time-period.

Clause 13: The method of any one or more of clauses 10-12, wherein the first sensing information comprises at least one of range or velocity information of a target object.

Clause 14: The method of any one or more of clauses 10-13, further comprising receiving, from each network entity of the first plurality of network entities and the second plurality of network entities, one or more of: a transmit power of the network entity, an effective isotropic radiated power of the network entity, a plurality of beams of the network entity, or direction information for each of the plurality of beams of the network entity.

Clause 15: The method of clause 14, further comprising determining the second grouping information based on the one or more of: the transmit power of the network entity, the effective isotropic radiated power of the network entity, the number of beams of the network entity, or the direction information for each of one or more beams of the network entity.

Clause 16: The method of any one or more of clauses 10-15, further comprising determining the second grouping information based on the first sensing information.

Clause 17: An apparatus, comprising: a memory comprising executable instructions; and a processor configured to execute the executable instructions and cause the apparatus to perform a method in accordance with any one of Clauses 1-16.

Clause 18: An apparatus, comprising means for performing a method in accordance with any one of Clauses 1-16.

Clause 19: A non-transitory computer-readable medium comprising executable instructions that, when executed by a processor of an apparatus, cause the apparatus to perform a method in accordance with any one of Clauses 1-16.

Clause 20: A computer program product embodied on a computer-readable storage medium comprising code for performing a method in accordance with any one of Clauses 1-16.

The preceding description is provided to enable any person skilled in the art to practice the various aspects described herein. The examples discussed herein are not limiting of the scope, applicability, or aspects set forth in the claims. Various modifications to these aspects will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to other aspects. For example, changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various actions may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method that is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

The various illustrative logical blocks, modules and circuits described in connection with the present disclosure may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any commercially available processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, a system on a chip (SoC), or any other such configuration.

As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-b-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

As used herein, the term “determining” encompasses a wide variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” may include resolving, selecting, choosing, establishing and the like.

The methods disclosed herein comprise one or more actions for achieving the methods. The method actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of actions is specified, the order and/or use of specific actions may be modified without departing from the scope of the claims. Further, the various operations of methods described above may be performed by any suitable means capable of performing the corresponding functions. The

means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor.

The following claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims. Within a claim, reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase “means for”. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

What is claimed is:

1. A method for wireless communications by a first network entity, the method comprising:

receiving, from a second network entity, first grouping information comprising network entity identifiers for a first plurality of network entities including the first network entity, the first grouping information associated with a first time-period;

receiving, from the second network entity, second grouping information comprising network entity identifiers for a second plurality of network entities including the first network entity, the second plurality of network entities being different than the first plurality of network entities, the second grouping information associated with a second time-period;

receiving a first radar signal, transmitted from at least one of the first plurality of network entities or at least one of the second plurality of network entities, during the first time-period;

receiving a second radar signal, transmitted from at least one of the second plurality of network entities or at least one of the first plurality of network entities, during the second time-period;

transmitting, to the second network entity, first sensing information based on the first radar signal; and transmitting, to the second network entity, second sensing information based on the second radar signal.

2. The method of claim 1, wherein the first grouping information comprises time-frequency resources for communication of radar signals by the first plurality of network entities.

3. The method of claim 1, wherein the first grouping information further comprises beam scanning directions for communication of radar signals by the first plurality of network entities.

4. The method of claim 1, further comprising transmitting, during the first time-period, a third radar signal according to the first grouping information.

5. The method of claim 1, further comprising transmitting, during the first time-period, the first radar signal according to the first grouping information.

6. The method of claim 1, wherein the first sensing information comprises at least one of range or velocity information of a target object.

7. The method of claim 1, further comprising transmitting, to the second network entity, one or more of: a transmit

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power of the first network entity, an effective isotropic radiated power of the first network entity, a plurality of beams of the first network entity, or direction information for each of the plurality beams of the first network entity.

8. A method for wireless communications by a first network entity, the method comprising:

transmitting, by the first network entity to a first plurality of network entities, first grouping information comprising network entity identifiers for the first plurality of network entities including a second network entity, the first grouping information associated with a first time-period, wherein the first grouping information comprises first time-frequency resources and beam scanning directions for communication of radar signals by the first plurality of network entities;

transmitting, by the first network entity to a second plurality of network entities, second grouping information comprising a network entity identifiers for the second plurality of network entities including the second network entity, the second plurality of network entities being different than the first plurality of network entities, the second grouping information associated with a second time-period, wherein the second grouping information comprises second time-frequency resources and beam scanning directions for communication of radar signals by the second plurality of network entities;

receiving first sensing information associated with the first time-period, wherein the first sensing information is based on a first radar signal transmitted to at least one of the first plurality of network entities or at least one of the second plurality of network entities; and

receiving second sensing information associated with the second time-period, wherein the second sensing information is based on a second radar signal transmitted to at least one of the first plurality of network entities or at least one of the second plurality of network entities.

9. The method of claim 8, wherein the first sensing information comprises at least one of range or velocity information of a target object.

10. The method of claim 8, further comprising receiving, from each network entity of the first plurality of network entities and the second plurality of network entities, one or more of: a transmit power of the network entity, an effective isotropic radiated power of the network entity, a plurality of beams of the network entity, or direction information for each of the plurality of beams of the network entity.

11. The method of claim 10, further comprising determining the second grouping information based on the one or more of: the transmit power of the network entity, the effective isotropic radiated power of the network entity, the number of beams of the network entity, or the direction information for each of one or more beams of the network entity.

12. The method of claim 8, further comprising determining the second grouping information based on the first sensing information.

13. A first network entity configured for wireless communications, comprising: a memory comprising computer-executable instructions; and a processor configured to execute the computer-executable instructions and cause the first network entity to:

receive, from a second network entity, first grouping information comprising network entity identifiers for a first plurality of network entities including the first network entity, the first grouping information associated with a first time-period;

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receive, from the second network entity, second grouping information comprising network entity identifiers for a second plurality of network entities including the first network entity, the second plurality of network entities different than the first plurality of network entities, the second grouping information associated with a second time-period;

receive a first radar signal, transmitted from at least one of the first plurality of network entities or at least one of the second plurality of network entities, during the first time-period;

receive a second radar signal, transmitted from at least one of the first plurality of network entities or at least one of the second plurality of network entities, during the second time-period;

transmit, to the second network entity, first sensing information based on the first radar signal; and transmit, to the second network entity, second sensing information based on the second radar signal.

14. The first network entity of claim 13, wherein the first grouping information comprises time-frequency resources for communication of radar signals by the first plurality of network entities.

15. The first network entity of claim 13, wherein the first grouping information further comprises beam scanning directions for communication of radar signals by the first plurality of network entities.

16. The first network entity of claim 13, wherein the processor is configured to execute the computer-executable instructions and further cause the first network entity to transmit, during the first time-period, a third radar signal according to the first grouping information.

17. The first network entity of claim 13, wherein the processor is configured to execute the computer-executable instructions and further cause the first network entity to transmit, during the first time-period, the first radar signal according to the first grouping information.

18. The first network entity of claim 13, wherein the first sensing information comprises at least one of range or velocity information of a target object.

19. The first network entity of claim 13, wherein the processor is configured to execute the computer-executable instructions and further cause the first network entity to transmit, to the second network entity, one or more of: a transmit power of the first network entity, an effective isotropic radiated power of the first network entity, a number of beams of the first network entity, or direction information for each of one or more beams of the first network entity.

20. A first network entity configured for wireless communications, comprising: a memory comprising computer-executable instructions; and a processor configured to execute the computer-executable instructions and cause the first network entity to:

transmit, by the first entity to a first plurality of network entities, first grouping information comprising network entity identifiers for the first plurality of network entities including a second network entity, the first grouping information associated with a first time-period, wherein the first grouping information comprises first time-frequency resources and beam scanning directions for communication of radar signals by the first plurality of network entities;

transmit, by the first network entity to a second plurality of network entities, second grouping information comprising a network entity identifiers for the second plurality of network entities including the second network entity, the second plurality of network entities

being different than the first plurality of network entities, the second grouping information associated with a second time-period, wherein the second grouping information comprises second time-frequency resources and beam scanning directions for communication of radar signals by the second plurality of network entities; 5
receive first sensing information associated with the first time-period, wherein the first sensing information is based on a first radar signal transmitted to at least one of the first plurality of network entities or at least one 10
of the second plurality of network entities; and
receive second sensing information associated with the second time-period, wherein the second sensing information is based on a second radar signal transmitted to at least one of the first plurality of network entities or 15
at least one of the second plurality of network entities.

21. The first network entity of claim 20, wherein the first sensing information comprises at least one of range or velocity information of a target object.

22. The first network entity of claim 20, wherein the 20
processor is configured to execute the computer-executable instructions and further cause the first network entity to receive, from each network entity of the first plurality of network entities and the second plurality of network entities, one or more of: a transmit power of the network entity, an 25
effective isotropic radiated power of the network entity, a plurality of beams of the network entity, or direction information for each of the plurality of beams of the network entity.

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